

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\V-310057%.pvdb

Date of Analysis : May 20, 2026 1:54pm

PV Elite 28, January 2026



Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 700.000 mm ID	
Vessel Design Length, Tangent to Tangent	2600.00 mm
Distance of Bottom Tangent above Grade	300.00 mm
Specified Datum Line Distance	0.00 mm
Internal Design Temperature	150 °C
Internal Design Pressure	0.350 MPa
External Design Temperature	150 °C
External Design Pressure	0.103 MPa
Maximum Allowable Working Pressure	1.66 MPa
External Max. Allowable Working Pressure	0.34 MPa
Shop Test Pressure	0.486 MPa
Field Test Pressure	2.161 MPa
Required Minimum Design Metal Temperature	11.4 °C
Warmest Computed Minimum Design Metal Temperature	-48.0 °C
Wind Design Code	IS 875 (PART 3) Amd. 1&2 (2020)
Earthquake Design Code	IS 1893 RSM Amd. 1&2 (2018)

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Nozzle	SA-106 B	...	...	K03006	
Nozzle	SA-105	...	...	K03504	
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-105	...	...	K03504	
Leg Baseplate	SA-516 70	...	...	K02700	
Leg Bolting	SA-193 B7	...	100 < t <= 180	G41400	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (MPa & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Bottom Head	0.359	0.10	2.550	1.5000	No	No
Shell-1	0.358	0.10	2.140	1.5000	N/A	No
Top Head	0.350	0.10	2.560	1.5000	No	No

Liquid Level: 1050.00 mm Dens.: 0.001 kg/cm³ Sp. Gr.: 0.916

Element Types and Properties:

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Ellipse	50.0	50.0	10.0	8.0	3.0	3.4	1.00	1.00
Cylinder	2550.0	2500.0	8.0	8.0	3.0	5.5	0.85	0.85
Ellipse	2600.0	50.0	10.0	8.0	3.0	3.4	1.00	1.00

Local Stress Analysis Results:



Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N3_2"	WRC-107/537	0.449	n/a	Passed
N1_6"	WRC-297	0.969	Shell	Passed
N4_2"	WRC-297	0.969	Shell	Passed
N2_4"	WRC-107/537	0.925	n/a	Passed

Loads for Foundation/Support Design:

Total Wind Shear on top of all Legs	5001.	N
Total Earthquake Shear on top of all Legs	1525.	N
Total Wind Moment at top of all Legs	6469526.	N-mm
Total Earthquake Moment at top of all Legs	2025672.	N-mm
Wind Load Case Test Moment on Support	2279542.	N-mm
Earthquake Load Case Test Moment on Support	215819.	N-mm
Test Case Wind Shear on Support	1650.	N
Max. Wind Shear on one Leg (top & bottom)	1667.	N
Max. Earthq. Shear on one Leg (top & bottom)	508.	N
Max. Wind Moment at base of one Leg	1967833.	N-mm
Max. Earthquake Moment at base of one Leg	600146.	N-mm
Max. Vertical Load (Wt. + Wind) on one Leg	21858.	N
Max. Vertical Load (Wt. + Eq.) on one Leg	9451.	N

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	776.7	kgm
Shop Test - Fabricated + water (full)	1866.6	kgm
Shipping - Fab. + rem. intls.+ shipping app.	862.4	kgm
Erected - Fab. + rem. intls.+ insul. (etc)	862.4	kgm
Empty - Fab. + intls. + details + wghts.	862.4	kgm
Operating - Empty + operating liquid (no ca)	1219.5	kgm
Field Test - Empty weight + test medium (full)	1952.4	kgm

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Nozzle Calculation Summary (MPa & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N3_2"	2.55	...	...	OK	3.00	OK	No Calc[*]
N1_6"	2.14	OK	...	OK	3.00	OK	Passed
K1A_1.5"	2.14	...	...	OK	3.00	OK	No Calc[*]
K1B_1.5"	2.14	...	...	OK	3.00	OK	No Calc[*]
H1_4"	1.66	OK	...	...		OK	Passed
H2_4"	1.66	OK	...	...		OK	Passed
N4_2"	2.14	...	...	OK	3.00	OK	No Calc[*]
N2_4"	1.86	OK	...	OK	3.00	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 1.66 Nozzle : H14"
 Minimum MAWP Shells/Flanges : 1.66 Element : Shell-1
 Minimum MAPnc Shells/Flanges : 2.64 Element : Shell-1

Computed Vessel M.A.W.P. : 1.66 MPa

[*] - This was a small opening and the areas were not computed or the MAWP of this connection could not be computed because the longitudinal bending stress was greater than the hoop stress.

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm	Layout Angle deg	Dia. Limit mm
10	N3_2"	0.000	0.000	107.600
20	N1_6"	1600.000	270.000	298.659
20	K1A_1.5"	1050.000	90.000	83.104
20	K1B_1.5"	150.000	90.000	83.104
20	H1_4"	600.000	270.000	204.641
20	H2_4"	2350.000	90.000	204.641
20	N4_2"	200.000	270.000	107.600
30	N2_4"	0.000	0.000	204.641

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
 lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
K1A_1.5"	1.500 in	Actual	LWN	2.760	16.002	...	...	192.71	300
K1B_1.5"	1.500 in	Actual	LWN	2.760	16.002	...	...	192.71	300
N3_2"	2.000 in	Actual	LWN	3.320	16.764	...	...	194.00	600
N4_2"	2.000 in	Actual	LWN	3.320	16.764	...	...	187.15	600
H1_4"	4.000 in	80	WNF	4.500	8.560	...	...	140.02	300
H2_4"	4.000 in	80	WNF	4.500	8.560	...	...	140.02	300
N2_4"	4.000 in	80	WNF	4.500	8.560	...	...	36.24	300
N1_6"	6.000 in	80	WNF	6.625	10.973	300.00	8.00	123.24	300

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
K1A_1.5"	SA-105	8.000	10.000	...	...	...
K1B_1.5"	SA-105	8.000	10.000	...	...	...
N3_2"	SA-106 B	8.000	10.000	...	...	...
N4_2"	SA-105	8.000	10.000	...	...	...
H1_4"	SA-106 B	8.000	10.000	...	...	...
H2_4"	SA-106 B	8.000	10.000	...	...	...
N2_4"	SA-106 B	8.000	10.000	...	...	...
N1_6"	SA-106 B	8.000	10.000	6.000	8.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
K1A_1.5"	1050.000	90.0	182.95	0.00	Shell-1
K1B_1.5"	150.000	90.0	182.95	0.00	Shell-1
N3_2"	...	0.0	184.60	0.00	Bottom Head
N4_2"	200.000	270.0	176.60	0.00	Shell-1
H1_4"	600.000	270.0	127.32	0.00	Shell-1
H2_4"	2350.000	90.0	127.32	0.00	Shell-1
N2_4"	...	0.0	25.67	0.00	Top Head
N1_6"	1600.000	270.0	104.97	0.00	Shell-1

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	ELLIPTICAL HEAD: 2.0 X 1, 10.0mm THK X 700.0mm ID X 50.0mm	SA-516 70
1	CYLINDER: 8.0mm THK X 700.0mm ID X 2500.0mm	SA-516 70
2	INSULATION: 225mm X 70mm THK	
1	INSULATION: 2500mm X 70mm THK	
3	LEGS, SHAPE: PIPE 1180.0mm	...
3	BASEPLATES: 150.0mm X 150.0mm X 10.0mm	SA-516 70
3	BASEPLATE BOLTS: 30.0mm	SA-193 B7
2	CLASS 300 GR 1.1, 4.0" BLIND FLANGE(S)	SA-105
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Bottom Head	[10]	B	-29	-48	-29	0.652	8.000	1.00	No
Bottom Head	[7]	B	-29	-48	-29	0.505	10.000	1.00	No
Shell-1	[8]	B	-29	-48	-29	0.660	8.000	0.85	No
Top Head	[10]	B	-29	-48	-29	0.648	8.000	1.00	No
Top Head	[7]	B	-29	-48	-29	0.502	10.000	1.00	No
N32"	[1]	B	-29	-104	-29	0.126	8.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N16"	[1]	B	-8	-104	-29	0.027	9.601	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K1A1.5"	[1]	A	-8	-104	-29	0.137	8.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K1B1.5"	[1]	A	-8	-104	-29	0.140	8.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
H14"	[1]	B	-8	-104	-29	0.026	7.490	1.00	No
Nozzle Flg	[5]	A	-18	-104					
H24"	[1]	B	-8	-104	-29	0.025	7.490	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N42"	[1]	A	-8	-104	-29	0.140	8.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N24"	[1]	B	-8	-104	-29	0.025	7.490	1.00	No
Nozzle Flg	[5]	A	-18	-104					
Nozl. Bolting[22]			-48			1.000	64.000	1.00	No

Warmest MDMT: -8 -48

Required Minimum Design Metal Temperature 11.4 °C
 Warmest Computed Minimum Design Metal Temperature -48.0 °C

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.
 UCS-66(g) was not considered.
 UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.
 UCS-66(i) applies to impact tested materials not by specification and
 UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
 The Basic MDMT includes the (30F) PWHT credit if applicable.



Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (SIASME):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	4.4480	N
Mass	Pounds	0.4536	kgm
Area	sq. inches	645.1600	mm ²
Moment	ft. lbs.	1356.3000	N-mm
Stress	lbs./sq.in.	0.0069	MPa
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0069	MPa
Elast. Modulus	lbs./sq.in.	0.0069	MPa
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kg/cm ²
Inertia	in.4	416231.0000	mm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	0.0479	kPa
Elevation	Feet	304.8000	mm
Volume	in.3	0.0164	ltr
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest)	0.35	MPa
Design Internal Temperature	150.0	°C
Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Minimum Design Metal Temperature	11.4	°C
Type of Construction	Welded	
Special Service	None	
Degree of Radiography	RT-1	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000



Consider Vortex Shedding

N

Shop Pressure Test:

Type of Pressure Test User Entered Pressure
 Pressure Test Position Horizontal
 Test Performed in Corroded Condition No

Field Pressure Test:

Type of Pressure Test UG-99(b)
 Pressure Test Position Vertical
 Test Performed in Corroded Condition Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	km/hr
Wind Zone Number	5	
Base Elevation	300	mm
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1] 1.0	
Terrain Category	2	
Equipment Class	Class A	
Topography Factor	[k3] 1.0	
Importance Factor	[k4] 1.15	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	

Seismic Design Code	IS 1893-RSM
Importance Factor for IS-1893	2.000
Soil Factor	1.000
Zone Number	2.000
Percent Seismic for Hydrotest	0.000
Period of Vibration (User defined) Sec	0.000
Damping Factor	2.000

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117



0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Consider Vertical Acceleration	Yes
Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N
Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/3
Element From Node	10
Element To Node	20
Element Type	Elliptical
Description	Bottom Head
Distance "FROM" to "TO"	50 mm
Inside Diameter	700 mm
Element Thickness	8 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	10 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	0.35 MPa
Design Temperature Internal Pressure	150 °C
Design External Pressure	0.1034 MPa
Design Temperature External Pressure	150 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	138 MPa
Allowable Stress, Operating	138 MPa
Allowable Stress, Hydrotest	234 MPa
Material Density	0.00775 kg/cm³
P Number Thickness	32.004 mm
Yield Stress, Operating	232 MPa
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0



Weld is pre-Heated	No
Element From Node	10
Detail Type	Liquid
Detail ID	Liquid: 10
Dist. from "FROM" Node / Offset dist	-150 mm
Height/Length of Liquid	200 mm
Liquid Density	0.0009148 kg/cm ³
Element From Node	10
Detail Type	Insulation
Detail ID	Ins: 10
Dist. from "FROM" Node / Offset dist	-175 mm
Height/Length of Insulation	225 mm
Thickness of Insulation	70 mm
Density	0.00015 kg/cm ³
Element From Node	10
Detail Type	Nozzle
Detail ID	N32"
Dist. from "FROM" Node / Offset dist	0 mm
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	94.096 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B
Element From Node	10
Detail Type	Weight
Detail ID	Vortex Breaker
Dist. from "FROM" Node / Offset dist	-100 mm
Miscellaneous Weight	0 N
Offset from Element Centerline	0 mm

Element#	2/3
Element From Node	20
Element To Node	30
Element Type	Cylinder
Description	Shell-1
Distance "FROM" to "TO"	2500 mm
Inside Diameter	700 mm
Element Thickness	8 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	8 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	0.35 MPa
Design Temperature Internal Pressure	150 °C
Design External Pressure	0.1034 MPa
Design Temperature External Pressure	150 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Weld is pre-Heated	No

Element From Node	20
Detail Type	Liquid
Detail ID	Liquid: 20
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	850 mm
Liquid Density	0.0009147 kg/cm ³
Element From Node	20
Detail Type	Insulation
Detail ID	Ins: 20



Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	2500	mm
Thickness of Insulation	70	mm
Density	0.00015	kg/cm ³

Element From Node	20	
Detail Type	Nozzle	
Detail ID	N16"	
Dist. from "FROM" Node / Offset dist	1550	mm
Nozzle Diameter	6	in.
Nozzle Schedule	80	
Nozzle Class	300	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	20	
Detail Type	Nozzle	
Detail ID	K1A1.5"	
Dist. from "FROM" Node / Offset dist	1000	mm
Nozzle Diameter	1.5	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	74.319	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	

Element From Node	20	
Detail Type	Nozzle	
Detail ID	K1B1.5"	
Dist. from "FROM" Node / Offset dist	100	mm
Nozzle Diameter	1.5	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	74.319	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	

Element From Node	20	
Detail Type	Nozzle	
Detail ID	H14"	
Dist. from "FROM" Node / Offset dist	550	mm
Nozzle Diameter	4	in.
Nozzle Schedule	80	
Nozzle Class	300	
Layout Angle	270.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	20	
Detail Type	Nozzle	
Detail ID	H24"	
Dist. from "FROM" Node / Offset dist	2300	mm
Nozzle Diameter	4	in.
Nozzle Schedule	80	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	



Element From Node	20
Detail Type	Nozzle
Detail ID	N42"
Dist. from "FROM" Node / Offset dist	150 mm
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	270.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	98.99 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105

Element From Node	20
Detail Type	Leg
Detail ID	LEGS
Dist. from "FROM" Node / Offset dist	0 mm
Diameter at Leg Centerline	716 mm
Leg Orientation	3
Number of Legs	3

Element From Node	20
Detail Type	Weight
Detail ID	N1-plate
Dist. from "FROM" Node / Offset dist	1550 mm
Miscellaneous Weight	98.06 N
Offset from Element Centerline	200 mm

Element From Node	20
Detail Type	Weight
Detail ID	Lifting Lug-1
Dist. from "FROM" Node / Offset dist	2400 mm
Miscellaneous Weight	98.06 N
Offset from Element Centerline	400 mm

Element From Node	20
Detail Type	Weight
Detail ID	Lifting Lug-2
Dist. from "FROM" Node / Offset dist	2400 mm
Miscellaneous Weight	98.06 N
Offset from Element Centerline	400 mm

Element From Node	20
Detail Type	Weight
Detail ID	INS CLEAT
Dist. from "FROM" Node / Offset dist	1250 mm
Miscellaneous Weight	539.33 N
Offset from Element Centerline	400 mm

Element#	3/3
Element From Node	30
Element To Node	40
Element Type	Elliptical
Description	Top Head
Distance "FROM" to "TO"	50 mm
Inside Diameter	700 mm
Element Thickness	8 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	10 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	0.35 MPa
Design Temperature Internal Pressure	150 °C
Design External Pressure	0.1034 MPa
Design Temperature External Pressure	150 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0



Elliptical Head Factor	2.0
Weld is pre-Heated	No
Element From Node	30
Detail Type	Insulation
Detail ID	Ins: 30
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	225 mm
Thickness of Insulation	70 mm
Density	0.00015 kg/cm ³
Element From Node	30
Detail Type	Nozzle
Detail ID	N24"
Dist. from "FROM" Node / Offset dist	0 mm
Nozzle Diameter	4 in.
Nozzle Schedule	80
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B



XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
Bottom Head		...	50	...	50
Shell-1		...	2550	...	2500
Top Head		...	2600	...	50

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd MPa	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) MPa	Element#
Bottom Head		0.3595	10	1.5	700	138	1
Shell-1		0.3577	8	1.5	700	117.3	2
Top Head		0.35	10	1.5	700	138	3

Element Required Thickness and MAWP :

From	To	Design Pressure MPa	M.A.W.P. Corroded MPa	M.A.P. New & Cold MPa	Minimum Thickness mm	Required Thickness mm	Element#
Bottom Head		0.35002	2.55	3.15	8	3	1
Shell-1		0.35002	2.14	2.64	8	3	2
Top Head		0.35002	2.56	3.15	8	3	3

Minimum 1.66 2.64

MAWP: 1.66 MPa, limited by: Nozzle Reinforcement.

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Bottom Head

Elliptical Head Design Information:					
Design Pressure	P	0.359 MPa	Int. Design Temperature	150 °C	
Material		SA-516 70	Ext. Design Temperature	150 °C	
Length		50.000 mm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	700.000 mm	Actual Stress	19.4 MPa	
Spec. Min. Thk t,e, or sa		8.0000 mm	Spec. Nominal Thk.	10.000 mm	
Surface Area		675035.3 mm ²	SG of Contents	0.916	
Element Volume		64.2 ltr	Weight of Contents	564.3 N	
Empty Weight		543.3 N	Operating Weight	1107.6 N	
			Comp. All. Ope. Stress	127.5 MPa	
			Aspect Ratio	2.00	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 8.750 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 4.375 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (0.36 \cdot 703 \cdot 0.99) / (2 \cdot 138 \cdot 1 - 0.2 \cdot 0.36)$$

$$= 0.9106 + 1.5000 = 2.4106 \text{ mm}$$

Note: The thickness required was less than the Code Minimum, therefore the Code Minimum value of 1.5000 mm per UG-16 will be used.

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.009 MPa

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 6.5) / (0.99 \cdot 703 + 0.2 \cdot 6.5)$$

$$= 2.56 - 0.0094 = 2.55 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 8) / (1 \cdot 700 + 0.2 \cdot 8)$$

$$= 3.15 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (0.36 \cdot (0.99 \cdot 703 + 0.2 \cdot 6.5)) / (2 \cdot 1 \cdot 6.5)$$

$$= 19.364 \text{ MPa}$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (0.36 \cdot 352) / (138 \cdot 1 - 0.6 \cdot 0.36) + 1.5$$

$$= 2.417 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.009 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 8.5) / (352 + 0.6 \cdot 8.5)$$

$$= 3.29 - 0.0094 = 3.28 \text{ MPa}$$



Factor K, corroded condition [Kcor]:
 $= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$
 $= (2 + (703 / (2 \cdot 176))^2) / 6$
 $= 0.994346$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 6.048 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 8, tr = 4.24, c = 1.5 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.65$, Temp. Reduction = 19 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 10, tr = 4.29, c = 1.5 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.5$, Temp. Reduction = 32 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1.671 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	11 °C
Governing Thk.	10.000 mm	Basic MDMT	-29 °C
Stress/Thk. Ratio	0.505	Temperature Difference	19 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 8.750 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 4.375 mm

EXTERNAL PRESSURE RESULTS for: Bottom Head

Elliptical Head From 10 to 20 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
6.500	716.00	110.15	0.0012609	91.91

MAEP = $B / (K_0 \cdot Do / t) = 91.9 / (0.9 \cdot 110) = 0.93$ MPa

Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
1.855	716.00	386.09	0.0003597	35.94

MAEP = $B / (K_0 \cdot Do / t) = 35.9 / (0.9 \cdot 386) = 0.1$ MPa



Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell-1

Cylindrical Shell Design Information:					
Design Pressure	P	0.358 MPa	Int. Design Temperature	150 °C	
Material		SA-516 70	Ext. Design Temperature	150 °C	
Length		2500.000 mm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	700.000 mm	Actual Stress	23.0 MPa	
Spec. Min. Thk t _e , or s _a		8.0000 mm	Spec. Nominal Thk.	8.000 mm	
Surface Area		5623450.5 mm²	SG of Contents	0.916	
Element Volume		962.3 ltr	Weight of Contents	2937.5 N	
Empty Weight		3380.9 N	Operating Weight	6318.5 N	
			Comp. All. Ope. Stress	127.5 MPa	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (0.36 \cdot 352) / (138 \cdot 0.85 - 0.6 \cdot 0.36)$$

$$= 1.0737 + 1.5000 = 2.5737 \text{ mm}$$

Note: The thickness required was less than the Code Minimum, therefore the Code Minimum value of 1.5000 mm per UG-16 will be used.

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.008 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 6.5) / (352 + 0.6 \cdot 6.5)$$

$$= 2.15 - 0.0076 = 2.14 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 8) / (350 + 0.6 \cdot 8)$$

$$= 2.64 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (0.36 \cdot (352 + 0.6 \cdot 6.5)) / (0.85 \cdot 6.5)$$

$$= 23.006 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 1.130 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 8, t_r = 5.05, c = 1.5 mm, E* = 0.85

Thickness Ratio = t_r * E* / (t_g - c) = 0.66, Temp. Reduction = 19 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1.670 MPa



UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	11 °C
Governing Thk.	8.000 mm	Basic MDMT	-29 °C
Stress/Thk. Ratio	0.660	Temperature Difference	19 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	85 %

EXTERNAL PRESSURE RESULTS for: Shell-1

Cylindrical Shell From 20 to 30 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
6.500	716.00	2716.67	110.15	3.7942	0.0002831	28.29

$$MAEP = (4*B)/(3*(Do/t)) = (4*28.3)/(3*110) = 0.34 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
3.995	716.00	2716.67	179.24	3.7942	0.0001392	13.90

$$MAEP = (4*B)/(3*(Do/t)) = (4*13.9)/(3*179) = 0.1 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
6.500	716.00	55332.27	110.15	50.0000	0.0000923	9.22

$$MAEP = (4*B)/(3*(Do/t)) = (4*9.22)/(3*110) = 0.11 \text{ MPa}$$

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Top Head

Elliptical Head Design Information:					
Design Pressure	P	0.350 MPa	Int. Design Temperature		150 °C
Material		SA-516 70	Ext. Design Temperature		150 °C
Length		50.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	700.000 mm	Actual Stress		18.9 MPa
Spec. Min. Thk t,e, or sa		8.0000 mm	Spec. Nominal Thk.		10.000 mm
Surface Area		675035.3 mm²	SG of Contents		0.000
Element Volume		64.2 ltr	Weight of Contents		0.0 N
Empty Weight		543.3 N	Operating Weight		543.3 N
			Comp. All. Ope. Stress		127.5 MPa
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 8.750 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 4.375 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (0.35 \cdot 703 \cdot 0.99) / (2 \cdot 138 \cdot 1 - 0.2 \cdot 0.35)$$

$$= 0.8867 + 1.5000 = 2.3867 \text{ mm}$$

Note: The thickness required was less than the Code Minimum, therefore the Code Minimum value of 1.5000 mm per UG-16 will be used.

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 6.5) / (0.99 \cdot 703 + 0.2 \cdot 6.5)$$

$$= 2.56 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 8) / (1 \cdot 700 + 0.2 \cdot 8)$$

$$= 3.15 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (0.35 \cdot (0.99 \cdot 703 + 0.2 \cdot 6.5)) / (2 \cdot 1 \cdot 6.5)$$

$$= 18.856 \text{ MPa}$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (0.35 \cdot 352) / (138 \cdot 1 - 0.6 \cdot 0.35) + 1.5$$

$$= 2.393 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 8.5) / (352 + 0.6 \cdot 8.5)$$

$$= 3.29 \text{ MPa}$$

Factor K, corroded condition [Kcor]:



$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$$

$$= (2 + (703 / (2 \cdot 176))^2) / 6$$

$$= 0.994346$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 6.048 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Govrn. thk, $t_g = 8$, $t_r = 4.21$, $c = 1.5$ mm, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.65$, Temp. Reduction = 20 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, $t_g = 10$, $t_r = 4.26$, $c = 1.5$ mm, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.5$, Temp. Reduction = 32 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1.662 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	11 °C
Governing Thk.	10.000 mm	Basic MDMT	-29 °C
Stress/Thk. Ratio	0.502	Temperature Difference	19 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 8.750 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 4.375 mm

EXTERNAL PRESSURE RESULTS for: Top Head

Elliptical Head From 30 to 40 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
6.500	716.00	110.15	0.0012609	91.91

$$MAEP = B / (K_0 \cdot Do / t) = 91.9 / (0.9 \cdot 110) = 0.93 \text{ MPa}$$

Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
1.855	716.00	386.09	0.0003597	35.94

$$MAEP = B / (K_0 \cdot Do / t) = 35.9 / (0.9 \cdot 386) = 0.1 \text{ MPa}$$



Hydrostatic Test Pressure Results:

UG99(b) = 1.30 * M.A.W.P. * Sa/S 2.161 MPa
 UG99b(b)(35) = 1.30 * Design Pres * Sa/S 0.455 MPa
 UG99(c) = 1.30 * M.A.P. - Head(Hyd) 3.431 MPa
 UG100 = 1.10 * M.A.W.P. * Sa/S 1.828 MPa
 PED = max(1.43*DP, 1.25*DP*ratio) 0.501 MPa
 PED&UG99(b)(35) = max(1.43*DP, 1.3*DP*ratio) 0.501 MPa
 App 27-4 = M.A.W.P. 1.662 MPa

User Defined Hydrostatic Test Pressure at High Point 0.486 MPa

Horizontal Test performed per: User Hydro Pressure

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
Bottom Head	21.6	234.0	0.092	0.49
Shell-1	25.7	234.0	0.110	0.49
Top Head	21.6	234.0	0.092	0.49

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N3_2"	Nozzle	118.00	118.00	1.000
N1_6"	Nozzle	118.00	118.00	1.000
N1_6"	Pad	138.00	138.00	1.000
K1A_1.5"	Nozzle	138.00	138.00	1.000
K1B_1.5"	Nozzle	138.00	138.00	1.000
H1_4"	Nozzle	118.00	118.00	1.000
H2_4"	Nozzle	118.00	118.00	1.000
N4_2"	Nozzle	138.00	138.00	1.000
N2_4"	Nozzle	118.00	118.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
Bottom Head	138.00	138.00	1.000
Shell-1	138.00	138.00	1.000
Top Head	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
N3_2"	1.04	216.00	0.005
N1_6"	3.59	216.00	0.017
K1A_1.5"	0.88	225.00	0.004
K1B_1.5"	0.88	225.00	0.004
H1_4"	3.57	216.00	0.017
H2_4"	3.57	216.00	0.017
N4_2"	1.04	225.00	0.005
N2_4"	3.57	216.00	0.017

Elements Suitable for Test Pressure.

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Shop Pressure Test Results: Step: 7 1:54pm May 20,2026



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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	2.161 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	0.455 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	3.412 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	1.828 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	0.501 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	0.501 MPa
App 27-4	= M.A.W.P.	1.662 MPa

UG-99(b), Test Pressure Calculation:
 = Test Factor • MAWP • Stress Ratio
 = 1.3 • 1.66 • 1
 = 2.161 MPa

Vertical Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
Bottom Head	117.9	234.0	0.504	2.19
Shell-1	140.7	234.0	0.601	2.19
Top Head	116.5	234.0	0.498	2.16

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N3_2"	Nozzle	118.00	118.00	1.000
N1_6"	Nozzle	118.00	118.00	1.000
N1_6"	Pad	138.00	138.00	1.000
K1A_1.5"	Nozzle	138.00	138.00	1.000
K1B_1.5"	Nozzle	138.00	138.00	1.000
H1_4"	Nozzle	118.00	118.00	1.000
H2_4"	Nozzle	118.00	118.00	1.000
N4_2"	Nozzle	138.00	138.00	1.000
N2_4"	Nozzle	118.00	118.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
Bottom Head	138.00	138.00	1.000
Shell-1	138.00	138.00	1.000
Top Head	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
N3_2"	4.63	216.00	0.021
N1_6"	15.90	216.00	0.074
K1A_1.5"	3.92	225.00	0.017
K1B_1.5"	3.92	225.00	0.017
H1_4"	15.81	216.00	0.073
H2_4"	15.81	216.00	0.073
N4_2"	4.63	225.00	0.021
N2_4"	15.64	216.00	0.072



Elements Suitable for Test Pressure.

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**External Pressure Calculation Results Summary:****ASME Code, Section VIII Division 1, 2025****External Pressure Calculations:**

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B MPa
10	20	No Calc	716	6.5	0.0012609	91.9104
20	30	2716.67	716	6.5	0.00028313	28.287
30	40	No Calc	716	6.5	0.0012609	91.9104

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure MPa	External M.A.W.P. MPa
10	20	8	3.3545	0.10343	0.93
20	30	8	5.49459	0.10343	0.34
30	40	8	3.3545	0.10343	0.93

Minimum

0.34

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required mm^4	Ring Inertia Available mm^4
10	20	No Calc	No Calc	No Calc	No Calc
20	30	2716.67	55332.3	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kgm	Element ID Volume ltr	Corroded Metal Wgt. kgm	Corroded ID Volume ltr	Extra due Misc % kgm
10	20	55.4052	64.1523	47.0945	64.8975	3.87837
20	30	344.781	962.285	280.727	970.551	24.1347
30	40	55.4052	64.1523	47.0945	64.8975	3.87837
Total		455	1090.59	374	1100.35	31

Weight of Details:

From	Type	Weight of Detail kgm	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset, Dtl. Cent. mm	Description
10	Liqd	57.5434	...	-75	...	Liquid: 10
10	Insl	10.4284	...	-62.5	...	Ins: 10
10	Nozl	10.2674	...	-313.304	...	N3_2"
10	Wght	...	...	-100	...	Vortex Breaker
20	Liqd	299.565	...	425	...	Liquid: 20
20	Insl	64.8948	...	1250	...	Ins: 20
20	Nozl	31.4889	-402.489	1550	...	N1_6"
20	Nozl	8.10947	441.478	1000	...	K1A_1.5"
20	Nozl	8.10947	441.478	100	...	K1B_1.5"
20	Nozl	29.1559	-413.665	550	...	H1_4"
20	Nozl	29.1559	413.665	2300	...	H2_4"
20	Nozl	10.8014	-438.303	150	...	N4_2"
20	Legs	62.78	...	-590	...	LEGS
20	Wght	10	-200	1550	...	N1-plate
20	Wght	10	...	2400	400	Lifting Lug-1
20	Wght	10	...	2400	-400	Lifting Lug-2
20	Wght	55	...	1250	-400	INS CLEAT
30	Insl	10.4284	...	112.5	...	Ins: 30
30	Nozl	14.3358	...	233.84	...	N2_4"

Total Weight of Each Detail Type:

Liquid	357.1
Insulation	85.8
Nozzles	141.4
Legs	62.8
Weights	85.0

Sum of the Detail Weights 732.1 kgm

Weight Summation Results: (kgm)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	487.5	487.5	487.5	487.5	487.5	487.5
Nozzles	141.4	141.4	141.4	141.4	141.4	141.4
Legs	62.8	62.8	62.8	62.8	62.8	62.8
Wld Weights	85.0	85.0	85.0	85.0	85.0	85.0
Insulation	...	...	85.8	85.8	85.8	85.8
Ope. Liquid	...	...	...	...	...	357.1
Test Liquid	...	1089.9	...	...	...	...
Totals	776.7	1866.6	862.4	862.4	862.4	1219.5

Field Installation Options:

Miscellaneous Weight Percent: 7.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the



weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	776.7 kgm
Shop Test Wt.	- Fabricated weight + water (full)	1866.6 kgm
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	862.4 kgm
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	862.4 kgm
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	862.4 kgm
Operating Wt.	- Empty weight + operating liq. uncorroded	1219.5 kgm
Field Test Wt.	- Empty weight + test medium (full)	1952.4 kgm

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	243.6 kgm
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	535.4 kgm
Mass of the Upper 1/3 of the Vertical Vessel (empty)	274.9 kgm

Surface Areas of Elements:

From	To	Outside Surface Area mm ²	Inside Surface Area mm ²
10	20	675035	641109
20	30	5623450	5497787
30	40	675035	641109
Total		6973521.0	6780005.0 mm ²
Total		7.0	6.8 m ²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kgm	Total. Ele. Oper. Wgt. kgm	Total. Ele. Hydro. Wgt. kgm	Total Dtl. Offset Mom. N-mm	Oper. Wgt. No Liquid kgm
10	20	79.9795	137.523	144.093	...	79.9795
20	Legs	...	...	...	100532	...
Legs	30	635.632	935.197	1597.33	...	635.632
30	40	84.0478	84.0478	148.161	...	84.0478

Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kgm	Cumulative Oper. Wgt. kgm	Cumulative Hydro. Wgt. kgm
10	20	...	...	...
20	Legs	-79.9795	-137.523	-144.093
Legs	30	719.68	1019.24	1745.49
30	40	84.0478	84.0478	148.161

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. N-mm	Cumulative Oper. Mom. N-mm	Cumulative Hydro. Mom. N-mm
10	20	...	...	...
20	Legs	100532	100532	100532
Legs	30	...	...	...
30	40	...	...	...

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Element and Detail Weights: Step: 10 1:54pm May 20,2026

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Nozzle Flange MAWP Results: (MPa & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
N3_2"	9.01	10.20	150	600	GR 1.1	5.668	7.848	9.011	7.848
N1_6"	4.51	5.10	150	300	GR 1.1	3.171	3.594	4.510	3.594
K1A_1.5"	4.51	5.10	150	300	GR 1.1	...	...	...	...
K1B_1.5"	4.51	5.10	150	300	GR 1.1	...	...	...	...
H1_4"	4.51	5.10	150	300	GR 1.1	...	...	...	...
H2_4"	4.51	5.10	150	300	GR 1.1	...	...	...	...
N4_2"	9.01	10.20	150	600	GR 1.1	5.668	7.848	9.011	7.848
N2_4"	4.51	5.10	150	300	GR 1.1	4.040	2.726	4.510	2.726
Min Rating	2.726	5.100, [for Core Elements]					...	...	...

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME N-mm	Axial Force FE N	MAWP PD MPa	Maximum Pressure Pmax MPa	Pass Fail
N3_2"	77.851	0.50	500000	1300	1.662	7.848	Pass
N1_6"	196.088	0.50	4500001	4000	1.662	3.594	Pass
N4_2"	77.851	0.50	500000	1300	1.662	7.848	Pass
N2_4"	138.176	0.50	2000000	2700	1.662	2.726	Pass

Evaluating Flange on Nozzle [N32"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$$

$$= 16(500000) + 4 \cdot \text{abs}(1300)77.851 \leq \pi \cdot 77.851^3 [(9.011 - 1.662) + .500 \cdot 9.011]$$

$$= 0.84050E+07 \leq 0.17577E+08 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N32" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(9.011, 0.500 \cdot 9.011 + 9.011 - 8404991.0 / (\pi \cdot 77.851^3))$$

$$= \min(9.011, 7.848)$$

$$= 7.848 \text{ MPa}$$

Evaluating Flange on Nozzle [N16"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$$

$$= 16(4500001) + 4 \cdot \text{abs}(4000)196.088 \leq \pi \cdot 196.088^3 [(4.510 - 1.662) + .500 \cdot 4.510]$$

$$= 0.75139E+08 \leq 0.12092E+09 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N16" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(4.510, 0.500 \cdot 4.510 + 4.510 - 75138680.0 / (\pi \cdot 196.088^3))$$

$$= \min(4.510, 3.594)$$

$$= 3.594 \text{ MPa}$$

Evaluating Flange on Nozzle [N42"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$$

$$= 16(500000) + 4 \cdot \text{abs}(1300)77.851 \leq \pi \cdot 77.851^3 [(9.011 - 1.662) + .500 \cdot 9.011]$$

$$= 0.84050E+07 \leq 0.17577E+08 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N42" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(9.011, 0.500 \cdot 9.011 + 9.011 - 8404991.0 / (\pi \cdot 77.851^3))$$



= min(9.011, 7.848)
= 7.848 MPa

Evaluating Flange on Nozzle [N24"]:

= $16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$
= $16(2000000) + 4 \cdot \text{abs}(2700) 138.176 \leq \pi \cdot 138.176^3 [(4.510 - 1.662) + .500 \cdot 4.510]$
= $0.33493E+08 \leq 0.42311E+08$ Check Passes

Maximum Pressure for Flange on Nozzle N24" [Pmax]:

= min(PR, $FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)$)
= min(4.510, $0.500 \cdot 4.510 + 4.510 - 33492926.0 / (\pi \cdot 138.176^3)$)
= min(4.510, 2.726)
= 2.726 MPa

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 18.0429 Hz.

The Natural Frequency for the Vessel (Ope...) is 15.3731 Hz.

The Natural Frequency for the Vessel (Filled) is 11.7818 Hz.

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Wind Load Calculation Results:

Wind Load Calculations per India Std. IS 875 (PART 3):2015, Amd. 1&2 (2020):

Input Values:

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	km/hr
Wind Zone Number	5	
Base Elevation	300	mm
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1]	1.0
Terrain Category		2
Equipment Class		Class A
Topography Factor	[k3]	1.0
Importance Factor	[k4]	1.15
Damping Factor (Beta) for Wind (Ope)		0.0100
Damping Factor (Beta) for Wind (Empty)		0.0000
Damping Factor (Beta) for Wind (Filled)		0.0000
Use Gust Response Factor (Dynamic Analysis)		No

Calculated Values:

Vessel Height to Diameter ratio: 3.333

Checking the requirement for dynamic effect of wind (9.1):

Vessel Operating Natural Frequency: 15.373 Hz

Note: Vessel Natural frequency is ≥ 1 Hz.

Note: Vessel height to diameter ratio (3.333) is ≤ 5 .

Note: Dynamic wind analysis is not required.

Design Wind Speed [Vz]:

$$\begin{aligned} &= \text{Basic Wind Speed} \cdot k_1 \cdot k_2 \cdot k_3 \cdot k_4 \quad (\text{for the 1st element}) \\ &= 50 \cdot 1 \cdot 1 \cdot 1 \cdot 1.15 \\ &= 57.496 \text{ meters/s} \end{aligned}$$

Design Wind Pressure per 4.5.2 [pd]:

$$\begin{aligned} &= \max(0.7 \cdot p_z, K_d \cdot K_a \cdot K_c \cdot 0.6 \cdot V_z^2) \\ &= \max(0.7 \cdot 1984, 1 \cdot 1 \cdot 1 \cdot 0.6 \cdot 57.5^2) \\ &= 1983.5 \text{ N/m}^2 \end{aligned}$$

Element Wind Load (operating condition):

$$\begin{aligned} &= \text{Wind Area} \cdot C_f \cdot p_d \\ &= 0.26 \cdot 0.74 \cdot 1984 \\ &= 377.2 \text{ N} \end{aligned}$$

From	Height mm	k1	k2	k3	Vz m/sec	Cf
10	472.24	1.0000	1.0000	1.0000	57.50	0.7444
20	1775.00	1.0000	1.0000	1.0000	57.50	0.7444
30	3155.76	1.0000	1.0000	1.0000	57.50	0.7444

Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

- Cf - Correction factor for natural frequency
- D - Average outer insulated diameter of vessel, mm
- Df - Damping Factor < 0.75 Unstable, > 0.95 Stable
- Dr - Average outer insulated of top half of vessel, mm
- f - Natural frequency of vibration (Hertz)



f1 - Natural frequency of bare vessel based on a unit value of $(D/L^2)(10^4)$
 L - Total height of structure mm
 Lc - Total length of conical section(s) of vessel, mm
 tb - Uncorroded plate thickness at bottom of vessel, mm
 V30 - Design Wind Speed provided by user, km/hr
 Vc - Critical wind velocity, km/hr
 Vw - Maximum wind speed at top of structure, km/hr
 W - Total corroded weight of structure, N
 Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. N
 Z - Maximum amplitude of vibration at top of vessel, mm
 D1 - Logarithmic decrement (taken as 0.03 for Welded Structures)
 Vp - Vib. Chance, $\leq 0.31416E-05$ (High); $0.31416E-05 < 0.39270E-05$ (Probable)
 P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5
 :: $0/2600 = 0$

:: $[D / L^2] \cdot 10^4 < 8$ (English Units, ft.)
 :: $[2.81/8.53^2] \cdot 10^4 = 386$ [Geometry Violation]

Compute the vibration possibility. If $V_p > 0.39270E-05$ no chance. [Vp]:

$= W / (L \cdot Dr^2)$
 $= 11112 / (2600 \cdot 809^2)$
 $= 0.65250E-05$

Since V_p is $> 0.39270E-05$ no further vibration analysis is required !

Wind Loads on Masses/Equipment/Piping

ID	Wind Area mm ²	Elevation mm	Pressure kPa	Force N
Vortex Breake	0.00	200.00	1.98	0.00
N1-plate	0.00	1900.00	1.98	0.00
Lifting Lug-1	0.00	2750.00	1.98	0.00
Lifting Lug-2	0.00	2750.00	1.98	0.00
INS CLEAT	0.00	1600.00	1.98	0.00

The Natural Frequency for the Vessel (Ope...) is 15.3731 Hz.

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area mm ²	Wind Pressure kPa	Element Wind Load N
10	20	472.237	1027.2	255471	1.98426	414.945
20	30	1775	1027.2	2568000	1.98426	4171.05
30	40	3155.76	1027.2	255471	1.98426	414.945

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Seismic Analysis Results per IS 1893 (Part 4): 2024

Input Data:

Importance Factor as Entered by User I 2.0000
Zone Number 2

Soil Type Rocky or Hard Soil
Response reduction factor (Force factor) R 2.0000
Time Period of Vibration (program or user) 0.0650 Seconds
Value of Sa/g from user Spectrum Data 0.1345
Percent Damping 2.0 %
Considering MCE for load determination No

Calculated results:

Horizontal Seismic Coefficient per 7.3.1 [Ah]:
 $= (Sa/g)_{ss} \cdot DampVal \cdot (I/R)$
 $= (0.13 \cdot 1) \cdot 1$
 $= 0.134$

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Actual Time Period = 0.065 secs
Computed Sa/g = 0.13

Note:

It was indicated to ignore the damping factor, assuming that the Site Spectrum specified is per the actual damping value of this vessel.
The value of "DampVal" has been set to 1 (not used).

Earthquake Element Load, for the first Element:
 $= \text{Earthquake Weight} \cdot Ah \cdot \text{Scalar}$
 $= 1349 \cdot 0.13 \cdot 1$
 $= 181.3 \text{ N}$

Vertical load per 8.4:
 $= 2/3 \cdot Ah \cdot W \cdot \text{Scalar}$
 $= 2/3 \cdot 0.13 \cdot 11343 \cdot 1$
 $= 1016.8 \text{ N}$

The Natural Frequency for the Vessel (Ope...) is 15.3731 Hz.

Earthquake Load Calculation:



From	To	Earthquake Height mm	Earthquake Weight N	Element Ope Load N	Element Emp Load N	Element Vertical Load N
10	20	25	1348.55	181.321	96.6946	-120.881
20	Legs	50	...	...	...	-120.881
Legs	30	1300	9170.54	1233.04	768.475	895.904
30	40	2575	824.173	110.815	101.613	73.877

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The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm	Cumulative Wind Shear N	Earthquake Shear N	Wind Bending N-mm	Earthquake Bending N-mm
10	20	130.764	...	...	...	...
20	Legs	...	414.945	181.321	54281.7	23719.8
Legs	30	1250	5000.94	1525.18	6253706	1809852
30	40	2630.76	414.945	110.815	54281.6	14496.5

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
Wind: 1.100; Earthquake: 1.000.

Note:

Review the Vessel Design Summary for the cumulative shear force and bending moment on the support.

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Wind Deflection Calculations:

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear N	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	...	2.7693	2.7693	0.0035203
20	Legs	414.945	2.7693	2.7693	0.0035203
Legs	30	5000.94	2.79117	2.83279	0.0035552
30	40	414.945	2.83367	2.83454	0.0035552

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed km/hr	2nd Crit. Wind Speed km/hr
10	20	283.476	1771.73
20	30	283.476	1771.73
30	40	283.476	1771.73

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)
Allowable deflection : 13.000 Actual deflection : 2.835 mm

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Longitudinal Stress Constants:

From	To	Metal Area New mm ²	Metal Area Corroded mm ²	Section Modulus New mm ³	Section Modulus Corroded mm ³
10	20	17794	14488.2	3114744	2546737
20	30	17794	14488.2	3114744	2546737
30	40	17794	14488.2	3114744	2546737

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Longitudinal Allowable Stresses:

From	To	Tensile MPa	Hydrotest Tensile MPa	Compressive MPa	Hydrotest Compressive MPa
10	20	165.597	280.795	-127.505	-127.809
20	Legs	140.757	238.676	-127.505	-127.809
Legs	30	140.757	238.676	-127.505	-127.809
30	40	165.597	280.795	-127.505	-127.809

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Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have also been computed in the corroded condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure MPa	Longitudinal Stress External Pressure MPa	Longitudinal Stress Hydrotest Pressure MPa
10	20	9.39401	-2.87426	57.9849
20	30	9.39401	-2.87426	57.9849
30	40	9.39401	-2.87426	57.9849

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty MPa	Wght. Str. Operating MPa	Wght. Str. Hydrotest MPa	Wght. Str. Emp. Mom. MPa	Wght. Str. Opr. Mom. MPa
10	20	...	...	...	...	...
20	Legs	0.054137	0.093087	0.097534	0.039462	0.039462
Legs	30	-0.48714	-0.48714	-0.48714	...	...
30	40	-0.05689	-0.05689	-0.05689	...	...

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. MPa	Bend. Str. Oper. Wind MPa	Bend. Str. Oper. Equ. MPa	Bend. Str. Hyd. Wind MPa	Bend. Str. Hyd. Equ. MPa
10	20	...	...	...	...	...
20	Legs	0.039462	0.021307	0.0093108	0.0070314	...
Legs	30	...	2.45479	0.71043	0.81008	...
30	40	...	0.021307	0.0056903	0.0070314	...

Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case MPa	Vortex Shedding Empty Case MPa	Vortex Shedding Test Case MPa	Earthquake Empty Case MPa
10	20	...	...	...	...
20	Legs	...	...	...	0.0049653
Legs	30	...	...	...	0.47723
30	40	...	...	...	0.0052178

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind MPa	Longitudinal Stress Y Forces Seismic MPa
10	20	...	-0.0083441
20	Legs	...	-0.0083441
Legs	30	...	0.061842
30	40	...	0.0050995

Longitudinal Stresses due to User Forces and Moments:



From	To	Wind For/Mom Corroded MPa	Earthquake For/Mom Corroded MPa	Wind For/Mom No Corrosion MPa	Earthquake For/Mom No Corrosion MPa
10	20	...	...	...	...
20	Legs	...	...	...	...
Legs	30	...	...	...	...
30	40	...	...	...	...

Bending Stresses due to Blast Load:

From	To	Bending Stress MPa
10	20	...
20	Legs	...
Legs	30	...
30	40	...

**Stress Combination Load Cases for Vertical Vessels:****Load Case Definition Key**

IP = Longitudinal Stress due to Internal Pressure
 EP = Longitudinal Stress due to External Pressure
 HP = Longitudinal Stress due to Hydrotest Pressure
 NP = No Pressure
 EW = Longitudinal Stress due to Weight (No Liquid)
 OW = Longitudinal Stress due to Weight (Operating)
 HW = Longitudinal Stress due to Weight (Hydrotest)
 WI = Bending Stress due to Wind Moment (Operating)
 EQ = Bending Stress due to Earthquake Moment (Operating)
 EE = Bending Stress due to Earthquake Moment (Empty)
 HI = Bending Stress due to Wind Moment (Hydrotest)
 HE = Bending Stress due to Earthquake Moment (Hydrotest)
 WE = Bending Stress due to Wind Moment (Empty) (no CA)
 WF = Bending Stress due to Wind Moment (Filled) (no CA)
 CW = Longitudinal Stress due to Weight (Empty) (no CA)
 VO = Bending Stress due to Vortex Shedding Loads (Ope)
 VE = Bending Stress due to Vortex Shedding Loads (Emp)
 VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
 FW = Axial Stress due to Vertical Forces for the Wind Case
 FS = Axial Stress due to Vertical Forces for the Seismic Case
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
 BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

An asterisk (*) in the final column denotes overstress.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	165.60	0.00	127.50	0.0000	-0.0000
20	0.11	140.76	-0.01	127.50	0.0008	0.0001
20	1.97	140.76	-2.94	127.50	0.0140	0.0231
30		165.60	-0.08	127.50		0.0006

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		165.60	-0.01	127.50		0.0001
20	0.09	140.76		127.50	0.0006	
20	0.05	140.76	-0.90	127.50	0.0004	0.0071
30		165.60	-0.06	127.50		0.0004

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	165.60	0.00	127.50	0.0000	-0.0000
20	0.15	140.76		127.50	0.0011	
20	1.97	140.76	-2.94	127.50	0.0140	0.0231
30		165.60	-0.08	127.50		0.0006

Analysis of Load Case 4 : NP+OW+EQ+FS+BS



From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		165.60	-0.01	127.50		0.0001
20	0.13	140.76		127.50	0.0009	
20	0.29	140.76	-1.14	127.50	0.0020	0.0089
30		165.60	-0.06	127.50		0.0005

Analysis of Load Case 5 : NP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	280.79	0.00	127.81	0.0000	-0.0000
20	0.14	238.68		127.81	0.0006	
20	0.32	238.68	-1.30	127.81	0.0014	0.0101
30		280.79	-0.06	127.81		0.0005

Analysis of Load Case 6 : NP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	280.79	0.00	127.81	0.0000	-0.0000
20	0.14	238.68		127.81	0.0006	
20		238.68	-0.49	127.81		0.0038
30		280.79	-0.06	127.81		0.0004

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	9.39	165.60		127.50	0.0567	
20	9.55	140.76		127.50	0.0678	
20	11.36	140.76		127.50	0.0807	
30	9.36	165.60		127.50	0.0565	

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	9.39	165.60		127.50	0.0567	
20	9.53	140.76		127.50	0.0677	
20	9.68	140.76		127.50	0.0688	
30	9.35	165.60		127.50	0.0564	

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		165.60	-2.87	127.50		0.0225
20		140.76	-2.84	127.50		0.0223
20		140.76	-5.82	127.50		0.0456
30		165.60	-2.95	127.50		0.0232

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		165.60	-2.88	127.50		0.0226
20		140.76	-2.84	127.50		0.0223
20		140.76	-4.01	127.50		0.0314
30		165.60	-2.93	127.50		0.0230

Analysis of Load Case 11 : HP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	57.98	280.79		127.81	0.2065	
20	58.13	238.68		127.81	0.2435	
20	58.31	238.68		127.81	0.2443	
30	57.94	280.79		127.81	0.2063	



Analysis of Load Case 12 : HP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	57.98	280.79		127.81	0.2065	
20	58.12	238.68		127.81	0.2435	
20	57.50	238.68		127.81	0.2409	
30	57.93	280.79		127.81	0.2063	

Analysis of Load Case 13 : IP+WE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	9.39	165.60		127.50	0.0567	
20	9.49	140.76		127.50	0.0674	
20	8.91	140.76		127.50	0.0633	
30	9.34	165.60		127.50	0.0564	

Analysis of Load Case 14 : IP+WF+CW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	9.39	165.60		127.50	0.0567	
20	9.47	140.76		127.50	0.0673	
20	9.00	140.76		127.50	0.0639	
30	9.35	165.60		127.50	0.0564	

Analysis of Load Case 15 : IP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	9.39	165.60		127.50	0.0567	
20	9.53	140.76		127.50	0.0677	
20	8.91	140.76		127.50	0.0633	
30	9.34	165.60		127.50	0.0564	

Analysis of Load Case 16 : IP+VE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	9.39	165.60		127.50	0.0567	
20	9.49	140.76		127.50	0.0674	
20	8.91	140.76		127.50	0.0633	
30	9.34	165.60		127.50	0.0564	

Analysis of Load Case 17 : NP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	165.60	0.00	127.50	0.0000	-0.0000
20	0.13	140.76		127.50	0.0009	
20		140.76	-0.49	127.50		0.0038
30		165.60	-0.06	127.50		0.0004

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	9.39	138.00		106.25	0.0680	
20	9.52	117.30		106.25	0.0811	
20	8.97	117.30		106.25	0.0765	
30	9.34	138.00		106.25	0.0677	

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		138.00	-2.88	106.25		0.0271
20		117.30	-2.83	106.25		0.0266



20	117.30	-3.30	106.25	0.0311
30	138.00	-2.93	106.25	0.0275

Analysis of Load Case 20 : BL+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	9.39	165.60		127.50	0.0567	
20	9.53	140.76		127.50	0.0677	
20	8.91	140.76		127.50	0.0633	
30	9.34	165.60		127.50	0.0564	

Absolute Maximum of the all of the Stress Ratio's 0.2443

Governing Element: Shell-1

Governing Load Case 11 : HP+HW+HI

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
Bottom Head	58.0	...	...	...	58.0	+ Tens
Bottom Head	58.0	...	...	...	...	- Comp
Shell-1	58.0	0.1	...	...	58.1	+ Tens
Shell-1	58.0	0.1	...	...	...	- Comp
Shell-1	58.0	-0.5	...	0.8	58.3	+ Tens
Shell-1	58.0	-0.5	...	-0.8	...	- Comp
Top Head	58.0	-0.1	...	...	57.9	+ Tens
Top Head	58.0	-0.1	...	...	...	- Comp



Shop/Field Installation Options :

Insulation is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Liquid	386.375 mm
Center of Gravity of the Insulation	1300.000 mm
Center of Gravity of the Nozzles	1307.947 mm
Center of Gravity of the Legs	-540.000 mm
Center of Gravity of the Added Weights (Operating)	1605.882 mm
Center of Gravity of the Added Weights (Empty)	1605.882 mm
Center of Gravity of Bare Shell New and Cold	1300.000 mm
Center of Gravity of Bare Shell Corroded	1300.000 mm
Vessel CG in the Operating Condition	934.093 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	1197.510 mm
Vessel CG in the Test Condition	1254.726 mm

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RESULTS FOR LEGS : Operating Case Description: LEGS

Legs attached to: Shell-1

Section Properties : Circular Steel Pipe: PIPE

USA AISC 1989 Steel Table

Overall Leg Length	1180.000	mm
Effective Leg Length	Leglen 1000.000	mm
Distance Leg Up Side of Vessel	0.000	mm
Number of Legs	Nleg 3	
Cross Sectional Area for PIPE	Aleg 1945.756	mm ²
Section Inertia (strong axis)	1620935.875	mm ⁴
Section Inertia (weak axis)	1620935.875	mm ⁴
Section Modulus (strong axis)	36466.531	mm ³
Section Modulus (weak axis)	36466.531	mm ³
Radius of Gyration (strong axis)	28.863	mm
Radius of Gyration (weak axis)	28.863	mm

Leg Orientation - Diagonal

Overturning Moment at top of Legs	6469525.5	N-mm
Total Weight Load at top of Legs	W 11383.5	N
Total Shear force at top of Legs	5000.9	N
Additional force in Leg due to Bracing	Fadd 0.0	N
Occasional Load Factor	Occfac 1.333	
Effective Leg End Condition Factor	k 1.000	
Pipe Leg Inside Diameter (Pid)	73.660	mm
Pipe Leg Outside Diameter (Pod)	88.900	mm

Note: The Legs are Not Cross Braced

The Leg Shear Force includes Wind, Seismic, and Applied Load Effects.

Maximum Shear at top of one Leg [Vleg]:
 $= (\max(\text{Wind, Seismic, Blast}) + \text{applied forces}) (I_{\max} / I_{\text{tot}})$
 $= (5001) (1620936 / 4862808)$
 $= 1666.98 \text{ N}$

Axial Compression, Leg furthest from the Neutral Axis [Sma]:
 $= (W / N_{\text{leg}} + (M_{\text{leg}} / (N_{\text{legm}} \cdot R_n))) / A_{\text{leg}}$
 $= (11384 / 3 + (6466949 / (1 \cdot 358))) / 1946$
 $= 11.23 \text{ MPa}$

Axial Compression, Leg closest to the Neutral Axis [Sva]:
 $= (W / N_{\text{leg}}) / A_{\text{leg}}$
 $= (11384 / 3) / 1946$
 $= 1.95 \text{ MPa}$

Allowable Compressive Stress for the Selected Leg ($KL/r < C_c$) [Sa]:
 $= \text{Occfac} (1 - ((kl/r)^2 / (2 \cdot C_c^2))) F_y / (5/3 + 3(kl/r) / (8 \cdot C_c) - (kl/r^3) / (8 \cdot C_c^3))$
 $= 1.33 (1 - (34.6)^2 / (2 \cdot 127^2)) 248 / (5/3 + 3(34.6) / (8 \cdot 127) - (34.6^3) / (8 \cdot 127^3))$
 $= 180.38 \text{ MPa}$

Bending at the Bottom of the Leg closest to the N.A. [S]:
 $= (V_{\text{leg}} \cdot \text{Leglen} / S_{\text{mdwa}})$
 $= (1667 \cdot 1000 / 36467)$
 $= 45.72 \text{ MPa}$

Allowable Bending Stress[Sb]:
 $= (0.6 \cdot F_y \cdot \text{Occfac})$
 $= (0.6 \cdot 248 \cdot 1.33)$
 $= 198.53 \text{ MPa}$

AISC Unity Check [Sc](must be $\leq$ to 1 and > 0) :
 $= (S_{\text{ma}} / S_{\text{a}}) + (0.85 \cdot S) / ((1 - S_{\text{ma}} / S_{\text{pex}}) \cdot S_{\text{b}})$
 $= (11.2 / 180) + (0.85 \cdot 45.7) / ((1 - 11.2 / 1163) \cdot 199)$
 $= 0.2599$



Bolting Size Requirement for Leg Baseplates :

Baseplate Material		SA-516 70
Baseplate Allowable Stress	SBA	137.90 MPa
Baseplate Length	B	150.0000 mm
Baseplate Width	D	150.0000 mm
Baseplate Thickness	BTHK	10.0000 mm
Leg Dimension Along Baseplate Length	d	88.9000 mm
Leg Dimension Along Baseplate Width	b	88.9000 mm
Bolt Circle Diameter	BCD	540.0000 mm
Bolt Material		SA-193 B7
Bolt Allowable Tensile Stress	STBA	137.30 MPa
Bolt Allowable Shear Stress	SBShear	107.87 MPa
Anchor Bolt Nominal Diameter	BOD	30.0000 mm
Number of Anchor Bolts in Tension per Leg	NB	0
Total Number of Anchors Bolt per Leg	NBT	1
Bolt Corrosion Allowance	bca	1.5000 mm
Ultimate 28-day Concrete Strength	FCPRIME	8.750 MPa

Leg Baseplate and Bolting Analysis, including Moments

Shear Stress in a Single Bolt [tb]:
 = Shear Force/(Bolt Area • Number of Bolts)
 = 5001/(391 • 3)
 = 4.3 MPa. Must be less than 107.9 MPa.

The bolt locations are outside the plate

Review notes about nozzle loadings and supports in the Warnings report.

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RESULTS FOR LEGS : HydroTest Case Description: LEGS

Legs attached to: Shell-1

Section Properties : Circular Steel Pipe: PIPE

USA AISC 1989 Steel Table

Overall Leg Length	1180.000	mm
Effective Leg Length	Leglen 1000.000	mm
Distance Leg Up Side of Vessel	0.000	mm
Number of Legs	Nleg 3	
Cross Sectional Area for PIPE	Aleg 1945.756	mm ²
Section Inertia (strong axis)	1620935.875	mm ⁴
Section Inertia (weak axis)	1620935.875	mm ⁴
Section Modulus (strong axis)	36466.531	mm ³
Section Modulus (weak axis)	36466.531	mm ³
Radius of Gyration (strong axis)	28.863	mm
Radius of Gyration (weak axis)	28.863	mm

Leg Orientation - Diagonal

Overturning Moment at top of Legs	2279542.5	N-mm
Total Weight Load at top of Legs	W 17728.6	N
Total Shear force at top of Legs	1650.3	N
Additional force in Leg due to Bracing	Fadd 0.0	N
Occasional Load Factor	Occfac 1.333	
Effective Leg End Condition Factor	k 1.000	
Pipe Leg Inside Diameter (Pid)	73.660	mm
Pipe Leg Outside Diameter (Pod)	88.900	mm

Note: The Legs are Not Cross Braced

The Leg Shear Force includes Wind, Seismic, and Applied Load Effects.

Maximum Shear at top of one Leg [Vleg]:
 $= (\max(\text{Wind, Seismic, Blast}) + \text{applied forces}) (I_{\max} / I_{\text{tot}})$
 $= (1650) (1620936 / 4862808)$
 $= 550.10 \text{ N}$

Axial Compression, Leg furthest from the Neutral Axis [Sma]:
 $= (W / N_{\text{leg}} + (M_{\text{leg}} / (N_{\text{legm}} \cdot R_n))) / A_{\text{leg}}$
 $= (17730 / 3 + (2278635 / (1 \cdot 358))) / 1946$
 $= 6.31 \text{ MPa}$

Axial Compression, Leg closest to the Neutral Axis [Sva]:
 $= (W / N_{\text{leg}}) / A_{\text{leg}}$
 $= (17729 / 3) / 1946$
 $= 3.04 \text{ MPa}$

Allowable Compressive Stress for the Selected Leg ($KL/r < C_c$) [Sa]:
 $= \text{Occfac} (1 - ((kl/r)^2 / (2 \cdot C_c^2))) F_y / (5/3 + 3(kl/r) / (8 \cdot C_c) - (kl/r^3) / (8 \cdot C_c^3))$
 $= 1.33 (1 - (34.6)^2 / (2 \cdot 127^2)) 248 / (5/3 + 3(34.6) / (8 \cdot 127) - (34.6^3) / (8 \cdot 127^3))$
 $= 180.38 \text{ MPa}$

Bending at the Bottom of the Leg closest to the N.A. [S]:
 $= (V_{\text{leg}} \cdot \text{Leglen} / S_{\text{mdwa}})$
 $= (550 \cdot 1000 / 36467)$
 $= 15.09 \text{ MPa}$

Allowable Bending Stress[Sb]:
 $= (0.6 \cdot F_y \cdot \text{Occfac})$
 $= (0.6 \cdot 248 \cdot 1.33)$
 $= 198.53 \text{ MPa}$

AISC Unity Check [Sc](must be $\leq$ to 1 and > 0) :
 $= (S_{\text{ma}} / S_{\text{a}}) + (0.85 \cdot S) / ((1 - S_{\text{ma}} / S_{\text{pex}}) \cdot S_{\text{b}})$
 $= (6.31 / 180) + (0.85 \cdot 15.1) / ((1 - 6.31 / 1163) \cdot 199)$
 $= 0.0999$



Bolting Size Requirement for Leg Baseplates :

Baseplate Material		SA-516 70
Baseplate Allowable Stress	SBA	137.90 MPa
Baseplate Length	B	150.0000 mm
Baseplate Width	D	150.0000 mm
Baseplate Thickness	BTHK	10.0000 mm
Leg Dimension Along Baseplate Length	d	88.9000 mm
Leg Dimension Along Baseplate Width	b	88.9000 mm
Bolt Circle Diameter	BCD	540.0000 mm
Bolt Material		SA-193 B7
Bolt Allowable Tensile Stress	STBA	137.30 MPa
Bolt Allowable Shear Stress	SBShear	107.87 MPa
Anchor Bolt Nominal Diameter	BOD	30.0000 mm
Number of Anchor Bolts in Tension per Leg	NB	0
Total Number of Anchors Bolt per Leg	NBT	1
Bolt Corrosion Allowance	bca	1.5000 mm
Ultimate 28-day Concrete Strength	FCPRIME	8.750 MPa

Leg Baseplate and Bolting Analysis, including Moments

Shear Stress in a Single Bolt [tb]:

= Shear Force/(Bolt Area • Number of Bolts)

= 5001/(391 • 3)

= 4.3 MPa. Must be less than 107.9 MPa.

The bolt locations are outside the plate

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: N32"

From: 10

Pressure for Reinforcement Calculations	P	0.3595	MPa
Temperature for Internal Pressure	Temp	150	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	150	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Elliptical Head	D	700.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	8.0000	mm
Head Internal Corrosion Allowance	c	1.5000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	184.6000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	8.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N32"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 0.36 \cdot 703 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 0.36)$
 $= 0.8209$ mm



Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R(\exp([P/(Sn \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 26.9(\exp([0.36/(118 \cdot 1)] - 1))$
 $= 0.0821 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3742 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	107.6000	mm
Parallel to Vessel Wall, opening length	d	53.8000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	16.2500	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N32".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.8742	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 2.4106	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 3.0000	mm
Wall Thickness, shell/head, external pressure	trb2	= 1.7620	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 6.3000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.3, \max(3, 3)]$
 $= 3.0000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.87, 3)$
 $= 3.0000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	10.0,	Allowable	:	118.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	285.0 MPa	Passed
Occasional	:	0.2,	Allowable	:	156.9 MPa	Passed
Shear	:	9.4,	Allowable	:	82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 8, tr = 0.82, c = 1.5 mm, E* = 1
 Thickness Ratio = $tr \cdot E*/(tg - c) = 0.13$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $0.36/10.20 = 0.035$



Weld Size Calculations, Description: N32"

Intermediate Calc. for nozzle/shell Welds Tmin 8.5000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	5.9500 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 1.3983 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 193.9983 mm

Input Echo, WRC107/537 Item 1, Description: N32" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	1.5000	mm
Vessel Diameter	Dv	1260.000	mm
Vessel Thickness	Tv	8.000	mm
Design Temperature	T1	150.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	MPa
Vessel Hot S.I. Allowable	Smh	138.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	1.5000	mm
Nozzle Diameter	Dn	50.800	mm
Nozzle Thickness	Tn	16.764	mm
Nozzle Material		SA-106 B	
Nozzle UNS Number		K03006	
Nozzle Cold S.I. Allowable	SNmc	118.00	MPa
Nozzle Hot S.I. Allowable	SNmh	118.00	MPa
Design Internal Pressure	Dp	0.359	MPa
Include Pressure Thrust		No	

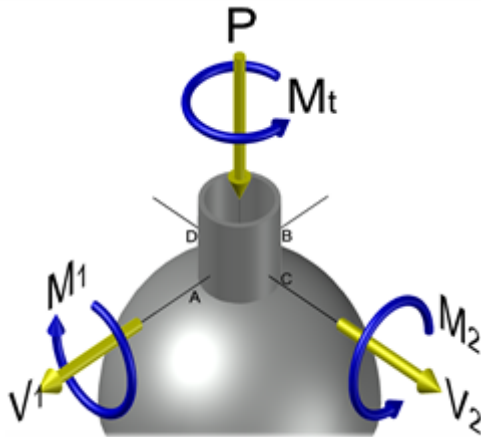
External Forces and Moments in WRC 107/537 Convention:

Radial Load	(SUS)	P	-1300.0	N
Longitudinal Shear	(SUS)	(Vl) V1	2000.0	N
Circumferential Shear	(SUS)	(Vc) V2	2000.0	N
Circumferential Moment	(SUS)	(Mc) M1	400000.0	N-mm
Longitudinal Moment	(SUS)	(Ml) M2	300000.0	N-mm
Torsional Moment	(SUS)	Mt	500000.0	N-mm

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 84.328 + 2 \cdot 1.65 \cdot \sqrt{(634.75 (8.0 - 1.5))}$$

$$= 296.297 \text{ mm}$$

WRC 107 Stress Calculation for SUSTAINED loads:

Radial Load	P	-1300.0	N
Circumferential Shear	(VC) V2	2000.0	N
Longitudinal Shear	(VL) V1	2000.0	N
Circumferential Moment	(MC) M1	400000.0	N-mm
Longitudinal Moment	(ML) M2	300000.0	N-mm
Torsional Moment	MT	500000.0	N-mm

Dimensionless Param: U = 0.66 TAU = 5.00 (2.26) RHO = 0.43

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 1	0.06054	(A,B,C,D)
$M(x) / P$	SP 1	0.07080	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.07710	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 1	0.15959	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.07710	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 1	0.15959	(A,B,C,D)
$N(y) * T / P$	SP 1	0.09642	(A,B,C,D)
$M(y) / P$	SP 1	0.02088	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.09187	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 1	0.05349	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.09187	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 1	0.05349	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb.	P	1.9	1.9	1.9	1.9	1.9	1.9	1.9	1.9
Rad. Bend.	P	13.1	-13.1	13.1	-13.1	13.1	-13.1	13.1	-13.1
Rad. Memb.	MC	0.0	0.0	0.0	0.0	-11.4	-11.4	11.4	11.4
Rad. Bend.	MC	0.0	0.0	0.0	0.0	-141.1	141.1	141.1	-141.1
Rad. Memb.	ML	-8.5	-8.5	8.5	8.5	0.0	0.0	0.0	0.0
Rad. Bend.	ML	-105.8	105.8	105.8	-105.8	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-99.4	86.1	129.3	-108.5	-137.5	118.5	167.4	-140.9

Tang. Memb. P	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0
Tang. Bend. P	3.9	-3.9	3.9	-3.9	3.9	-3.9	3.9	-3.9
Tang. Memb. MC	0.0	0.0	0.0	0.0	-13.5	-13.5	13.5	13.5
Tang. Bend. MC	0.0	0.0	0.0	0.0	-47.3	47.3	47.3	-47.3
Tang. Memb. ML	-10.2	-10.2	10.2	10.2	0.0	0.0	0.0	0.0
Tang. Bend. ML	-35.5	35.5	35.5	-35.5	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-38.8	24.4	52.4	-26.2	-54.0	32.9	67.6	-34.6
Shear VC	2.3	2.3	-2.3	-2.3	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-2.3	-2.3	2.3	2.3
Shear MT	6.9	6.9	6.9	6.9	6.9	6.9	6.9	6.9
Tot. Shear	9.2	9.2	4.6	4.6	4.6	4.6	9.2	9.2
Str. Int.	100.8	87.4	129.5	108.8	137.8	118.8	168.2	141.7

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	17.5	17.5	17.5	17.5	17.5	17.5	17.5	17.5
Rad.	Pl (SUS)	-6.7	-6.7	10.4	10.4	-9.5	-9.5	13.2	13.2
Rad.	Q (SUS)	-92.7	92.7	118.9	-118.9	-128.0	128.0	154.2	-154.2
Long.	Pm (SUS)	17.5	17.5	17.5	17.5	17.5	17.5	17.5	17.5
Long.	Pl (SUS)	-7.2	-7.2	13.1	13.1	-10.6	-10.6	16.5	16.5
Long.	Q (SUS)	-31.6	31.6	39.3	-39.3	-43.4	43.4	51.1	-51.1
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	2.3	2.3	-2.3	-2.3	-2.3	-2.3	2.3	2.3
Shear	Q (SUS)	6.9	6.9	6.9	6.9	6.9	6.9	6.9	6.9
Pm (SUS)		17.5	17.5	17.5	17.5	17.5	17.5	17.5	17.5
Pm+Pl (SUS)		12.9	12.9	31.9	31.9	9.8	9.8	35.2	35.2
Pm+Pl+Q (Total)		83.3	104.9	147.0	91.3	120.3	136.2	185.7	124.3

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	17.46	138.00	Passed
Pm+Pl (SUS)	35.17	207.00	Passed
Pm+Pl+Q (TOTAL)	185.69	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: N16"

From: 20

Pressure for Reinforcement Calculations	P	0.3500	MPa
Temperature for Internal Pressure	Temp	150	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	700.00	mm
Design Length of Section	L	2716.6667	mm
Shell Finished (Minimum) Thickness	t	8.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		1600.00	mm
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User Entered Minimum Design Metal Temperature		11.40	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		6.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		80	

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	104.9720	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	8.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	300.0000	mm
Thickness of Pad	te	8.0000	mm
Weld leg size between Pad and Shell	Wp	6.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	8.0000	mm
Reinforcing Pad Width		65.8625	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N16"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation		5.761	in.
--	--	-------	-----



Actual Thickness Used in Calculation 0.432 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.35 \cdot 352 / (138 \cdot 1 - 0.6 \cdot 0.35)$
 $= 0.8929$ mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.35 \cdot 74.7 / (118 \cdot 1 - 0.6 \cdot 0.35)$
 $= 0.2219$ mm

Required Nozzle thickness under External Pressure per UG-28 : 0.4432 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 298.6588 mm
 Parallel to Vessel Wall, opening length d 149.3294 mm
 Normal to Vessel Wall (Thickness Limit), pad side Tlwp 16.2500 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	135.790	303.739	NA
Area in Shell	A1	821.907	367.252	NA
Area in Nozzle Wall	A2	257.082	250.931	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	82.889	82.889	NA
Area in Element	A5	1043.070	1043.070	NA
TOTAL AREA AVAILABLE	Atot	2204.948	1744.142	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.
 The area available with the given pad is Sufficient.

Area Required [A]:
 $= 0.5 (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F (1 - fr1))$ per UG-37(d)
 $= 0.5 (149 \cdot 3.99 \cdot 1 + 2 \cdot 9.47 \cdot 3.99 \cdot 1 (1 - 0.86))$
 $= 303.739$ mm²

Reinforcement Areas per Figure UG-37.1



Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 149 (1 \cdot 6.5 - 1 \cdot 3.99) - 2 \cdot 9.47$$

$$(1 \cdot 6.5 - 1 \cdot 3.99) \cdot (1 - 0.86)$$

$$= 367.252 \text{ mm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$$

$$= (2 \cdot 16.2) \cdot (9.47 - 0.44) \cdot 0.86$$

$$= 250.931 \text{ mm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= (Wo^2 - Ar \text{ Lost}) \cdot Fr3 + ((Wi - can/0.707)^2 - Ar \text{ Lost}) \cdot fr2 + Wp^2 \cdot fr4$$

$$= (96.9) \cdot 0.86 + (0) \cdot 0.86 + 0^2 \cdot 1$$

$$= 82.889 \text{ mm}^2$$

Area Available in Element [A5]:

$$= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4$$

$$= (299 - 168) \cdot 8 \cdot 1$$

$$= 1043.070 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 1.9432 mm
Wall Thickness per UG16(b),	tr16b = 3.0000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 2.3929 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 3.0000 mm
Wall Thickness, shell/head, external pressure	trb2 = 1.7636 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm
Wall Thickness per table UG-45	tb3 = 7.7200 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[7.72, \max(3, 3)]$$

$$= 3.0000 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(1.94, 3)$$

$$= 3.0000 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 \cdot 10.973 = 9.601 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 25.8,	Allowable	: 118.0 MPa	Passed
Expansion	: 0.0,	Allowable	: 269.2 MPa	Passed
Occasional	: 1.3,	Allowable	: 156.9 MPa	Passed
Shear	: 17.4,	Allowable	: 82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 9.6, tr = 0.22, c = 1.5 mm, E* = 1
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.027$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 8, c = 1.5 mm, E* = 1
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.14$, Temp. Reduction = 78 °C
 Attachment governing, Attachment stress = Shell stress per Code



Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 8$, $c = 1.5$ mm, $E^* = 1$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.14$, Temp. Reduction = 78 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 8$, $t_r = 0.89$, $c = 1.5$ mm, $E^* = 1$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.14$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $t_g = 8$, $t_r = 0.89$, $c = 1.5$ mm, $E^* = 1$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.14$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -104 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $0.35/5.10 = 0.069$

Weld Size Calculations, Description: N16"

Intermediate Calc. for nozzle/shell Welds T_{min} 8.0000 mm
 Intermediate Calc. for pad/shell Welds T_{minPad} 6.5000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.6000 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm
Pad Weld	$3.2500 = 0.5 * T_{minPad}$	$4.2420 = 0.7 * W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (304 - 367 + 2 \cdot 9.47 \cdot 0.86 \cdot (1 \cdot 6.5 - 3.99)) \cdot 138)$
 $= \max(0, -3163.49) \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-Can/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (251 + 1043 + 82.9 - 0 \cdot 0.86) \cdot 138$
 $= 189991.25 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (251 + 0 + 85.5 + (105)) \cdot 138$
 $= 60953.49 \text{ N}$



Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$

$$= (251 + 0 + 82.9 + 1043 + (105)) \cdot 138$$

$$= 204521.03 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.14/2.0) \cdot 168 \cdot 10 \cdot 0.49 \cdot 118$$

$$= 152817. \text{ N}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.14/2.0) \cdot 300 \cdot 6 \cdot 0.49 \cdot 138$$

$$= 191171. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 79.4) \cdot (11 - 1.5) \cdot 0.7 \cdot 118$$

$$= 195160. \text{ N}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.14/2) \cdot 168 \cdot 8 \cdot 0.74 \cdot 138$$

$$= 215921. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 168 \cdot (8 - 1.5) \cdot 0.74 \cdot 138$$

$$= 175436. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (191171 + 195160) = 386331 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (152817 + 215921 + 175436 + 0) = 544175 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (191171 + 175436 + 0) = 366607 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 386331 N	, must exceed W = 0 N	or W1 = 189991 N
Path 2-2 = 544174 N	, must exceed W = 0 N	or W2 = 60953 N
Path 3-3 = 366607 N	, must exceed W = 0 N	or W3 = 204521 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2.14 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 10.2635 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 123.2355 mm

Input Echo, WRC297 Item 1, Description: N16" :

Diameter Basis for Cylindrical Shell	ID
Shell Internal Corrosion Allowance	1.5000 mm
Shell External Corrosion Allowance	0.0000 mm
Shell Diameter	700.000 mm
Shell Thickness	8.0000 mm



Shell Stress Concentration Factor		1.000	
Vessel Material		SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00	MPa
Vessel Hot S.I. Allowable	Smh	138.00	MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle		ID	
Nozzle Corrosion Allowance		1.5000	mm
Nozzle Diameter		146.329	mm
Nozzle Thickness		10.9728	mm
Nozzle Stress Concentration Factor		1.000	
Nozzle Material		SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	118.00	MPa
Nozzle Hot S.I. Allowable	SNmh	118.00	MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad		300.000	mm
Thickness of Reinforcing Pad		8.0000	mm

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUStained loads.*

Design Internal Pressure	Dp	0.35	MPa
Radial Load	P	-4000.00	N
Circumferential Shear	Vc	6000.00	N
Longitudinal Shear	Vl	6000.00	N
Circumferential Moment	Mc	0.270E+07	N-mm
Longitudinal Moment	Ml	0.360E+07	N-mm
Torsional Moment	Mt	0.450E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 168.275 + 2 \cdot 1.65 \cdot \sqrt{(358.75(8.0 - 1.5))}$
 $= 327.630 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	717.500	mm
Nozzle Outside Diameter	(d) =	168.275	mm
Vessel Thickness used	(T) =	14.500	mm
Nozzle Thickness used	(t) =	9.473	mm
T / t	=	1.531	
d / t	=	17.764	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.650$$

Nr/P	=	0.082
Mr/P	=	0.115
M0/P	=	0.051
N0/P	=	0.134
MrD/Mc	=	0.230
NrDL/Mc	=	0.131
M0d/Mc	=	0.120
N0DL/Mc	=	0.106
MrD/Ml	=	0.142
NrDL/Ml	=	0.087
M0D/Ml	=	0.074
N0DL/Ml	=	0.203



Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-20	-20	20	20
Outplane Bending (ML)	-45	45	45	-45
Normal Pressure Stress	8	8	8	8
Outplane Stress Summary	-49	30	82	-19

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	13	-13	13	-13
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-8	-8	8	8
Inplane Bending (ML)	-86	86	86	-86
Inplane Pressure Stress	4	4	4	4
Inplane Stress Summary	-76	70	114	-85

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	6	6	6	6
Shear Stress Summary	8	8	5	5

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	79	72	115	85

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	-8	-8	8	8
Outplane Bending (Mc)	-54	54	54	-54
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	4	4	4	4
Outplane Stress Summary	-50	47	75	-45

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	1	1	1	1



Inplane Bending (P)	13	-13	13	-13
Inplane Membrane (Mc)	-9	-9	9	9
Inplane Bending (Mc)	-105	105	105	-105
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	8	8	8	8

Inplane Stress Summary	-92	92	138	-98
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	6	6	6	6

Shear Stress Summary	5	5	8	8
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	93	93	139	99
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-20	-20	20	20
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	3	3	3	3

Hoop Stress Summary	-14	-14	26	26
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	19	-19	19	-19
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-20	-20	20	20
Axial Bending (ML)	-140	140	140	-140
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	-138	102	183	-137
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	10	10	10	10

Shear Stress Summary	13	13	8	8
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Outside	Inside	Outside	Inside



Two * Max Shear Stress 140 120 183 165

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Stresses in the
 hoop direction)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-8	-8	8	8
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	3	3	3	3

Hoop Stress Summary -2 -1 13 14

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Stresses Normal to
 pipe cross-section)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Axial Membrane (P)	0	0	0	0
Axial Bending (P)	19	-19	19	-19
Axial Membrane (Mc)	-15	-15	15	15
Axial Bending (Mc)	-176	176	176	-176
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1

Axial Stress Summary -169 144 214 -178

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Shear stress)

	Cu Outside	Cl Inside	Du Outside	Dl Inside
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Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	10	10	10	10

Shear Stress Summary 8 8 13 13

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Stress Intensities)

	Cu Outside	Cl Inside	Du Outside	Dl Inside
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Two * Max Shear Stress 170 146 214 194

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{354.75(8.0 - 1.5)}$$

$$= 326.739 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	709.500 mm
Nozzle Outside Diameter	(d) =	300.000 mm
Vessel Thickness used	(T) =	6.500 mm
Nozzle Thickness used	(t) =	75.335 mm
T / t	=	0.086
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.418$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.



Nr/P = 0.049
 Mr/P = 0.076
 M0/P = 0.022
 N0/P = 0.030
 MrD/Mc = 0.247
 NrDL/Mc = 0.102
 M0d/Mc = 0.074
 N0DL/Mc = 0.056
 MrD/Ml = 0.056
 NrDL/Ml = 0.055
 M0D/Ml = 0.017
 N0DL/Ml = 0.032

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	12	-12	12	-12
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-9	-9	9	9
Outplane Bending (ML)	-28	28	28	-28
Normal Pressure Stress	18	19	18	19
Outplane Stress Summary	-3	29	71	-10

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	43	-43	43	-43
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-15	-15	15	15
Inplane Bending (ML)	-94	94	94	-94
Inplane Pressure Stress	9	9	9	9
Inplane Stress Summary	-52	49	167	-108

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	6	6	2	2

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	53	51	167	108

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	12	-12	12	-12
Outplane Membrane (Mc)	-11	-11	11	11
Outplane Bending (Mc)	-94	94	94	-94

Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	9	9	9	9

Outplane Stress Summary	-81	82	130	-82
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	43	-43	43	-43
Inplane Membrane (Mc)	-21	-21	21	21
Inplane Bending (Mc)	-315	315	315	-315
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	18	19	18	19

Inplane Stress Summary	-270	274	404	-313
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4

Shear Stress Summary	2	2	6	6
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	270	274	404	313

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	8	8	8	8	8	8	8	8
Circ. Pl (SUS)	-18	-18	22	22	-8	-8	10	10
Circ. Q (SUS)	-40	40	50	-50	-92	92	118	-118
Long. Pm (SUS)	4	4	4	4	4	4	4	4
Long. Pl (SUS)	-7	-7	9	9	-6	-6	10	10
Long. Q (SUS)	-73	73	99	-99	-49	49	59	-59
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	6	6	6	6	6	6	6	6
Pm (SUS)	8.0	8.0	8.0	8.0	8.0	8.0	8.0	8.0
Pm+Pl (SUS)	10.1	10.1	30.1	30.1	2.8	2.8	18.2	18.2
Pm+Pl+Q (Total)	77.8	71.2	112.8	86.4	92.6	92.5	136.8	100.9

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
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Pm (SUS)	8.00	138.00	Passed
Pm+Pl (SUS)	30.06	207.00	Passed
Pm+Pl+Q (TOTAL)	136.77	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	18	19	18	19	18	19	18	19
Circ. Pl (SUS)	-7	-7	11	11	-17	-17	25	25
Circ. Q (SUS)	-16	16	40	-40	-272	272	358	-358
Long. Pm (SUS)	9	9	9	9	9	9	9	9
Long. Pl (SUS)	-11	-11	19	19	-9	-9	13	13
Long. Q (SUS)	-51	51	137	-137	-82	82	106	-106
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	18.0	19.0	18.0	19.0	18.0	19.0	18.0	19.0
Pm+Pl (SUS)	13.2	14.1	29.6	30.4	2.2	2.8	43.0	44.0
Pm+Pl+Q (Total)	53.5	50.1	165.1	109.1	271.0	274.0	401.1	314.1

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	19.00	138.00	Passed
Pm+Pl (SUS)	44.05	207.00	Passed
Pm+Pl+Q (TOTAL)	401.09	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	3	3	3	3	3	3	3	3
Circ. Pl (SUS)	-18	-18	22	22	-6	-6	10	10
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	-20	-20	20	20	-15	-15	15	15
Long. Q (SUS)	-121	121	159	-159	-157	157	195	-195
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	10	10	10	10	10	10	10	10
Pm (SUS)	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0
Pm+Pl (SUS)	19.8	19.8	25.8	25.8	14.4	14.4	17.0	17.0
Pm+Pl+Q (Total)	141.1	119.4	180.4	163.8	171.4	146.9	211.7	193.5



Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	3.00	118.00	Passed
Pm+Pl (SUS)	25.83	177.00	Passed
Pm+Pl+Q (TOTAL)	211.72	353.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: K1A1.5"

From: 20

Pressure for Reinforcement Calculations	P	0.3500	MPa
Temperature for Internal Pressure	Temp	150	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	700.00	mm
Design Length of Section	L	2716.6667	mm
Shell Finished (Minimum) Thickness	t	8.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		1050.00	mm
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User Entered Minimum Design Metal Temperature		11.40	°C
---	--	-------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		1.5000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0020	mm

Flange Material		SA-105	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	182.9500	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	8.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: K1A1.5"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation		1.500	in.
Actual Thickness Used in Calculation		0.630	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.35 \cdot 352 / (138 \cdot 1 - 0.6 \cdot 0.35)$
 $= 0.8929$ mm



Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= $R \left(\exp\left[\frac{P}{(S_n \cdot E)}\right] - 1 \right)$ per Appendix 1-2 (a)(1)
= $20.5 \left(\exp\left[\frac{0.35}{138 \cdot 1}\right] - 1 \right)$
= 0.0522 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.3356 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	83.1040	mm
Parallel to Vessel Wall	Rn+tn+t	41.5520	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	16.2500	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K1A1.5".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.8356	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 2.3929	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 3.0000	mm
Wall Thickness, shell/head, external pressure	trb2	= 1.7636	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 6.0200	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[6.02, max(3, 3)]
= 3.0000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(1.84, 3)
= 3.0000 mm

Available Nozzle Neck Thickness = 16.0020 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 8, tr = 0.89, c = 1.5 mm, E* = 1
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.14$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $0.35/5.10 = 0.069$

Weld Size Calculations, Description: K1A1.5"

Intermediate Calc. for nozzle/shell Welds	Tmin	6.5000	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.5500 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles

PV Elite 28 Licensee: SL User
FileName : V-31005_7%
Nozzle Calcs.: K1A_1.5"

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HEXAGON

exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 1.7596 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 192.7096 mm

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Input, Nozzle Desc: K1B1.5"

From: 20

Pressure for Reinforcement Calculations	P	0.3568	MPa
Temperature for Internal Pressure	Temp	150	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	700.00	mm
Design Length of Section	L	2716.6667	mm
Shell Finished (Minimum) Thickness	t	8.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		150.00	mm
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User Entered Minimum Design Metal Temperature		11.40	°C
---	--	-------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		1.5000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0020	mm

Flange Material		SA-105	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	182.9500	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	8.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: K1B1.5"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation		1.500	in.
Actual Thickness Used in Calculation		0.630	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.36 \cdot 352 / (138 \cdot 1 - 0.6 \cdot 0.36)$
 $= 0.9101 \text{ mm}$



Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R(\exp([P/(Sn \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 20.5(\exp([0.36/(138 \cdot 1)] - 1))$
 $= 0.0532 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3356 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	83.1040	mm
Parallel to Vessel Wall	Rn+tn+t	41.5520	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	16.2500	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K1B1.5".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.8356	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 2.4101	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 3.0000	mm
Wall Thickness, shell/head, external pressure	trb2	= 1.7636	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 6.0200	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.02, \max(3, 3)]$
 $= 3.0000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.84, 3)$
 $= 3.0000 \text{ mm}$

Available Nozzle Neck Thickness = 16.0020 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 8, tr = 0.91, c = 1.5 mm, E* = 1
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.14$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $0.36/5.10 = 0.070$

Weld Size Calculations, Description: K1B1.5"

Intermediate Calc. for nozzle/shell Welds	Tmin	6.5000	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.5500 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles

PV Elite 28 Licensee: SL User
FileName : V-31005_7%
Nozzle Calcs.: K1B_1.5"

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HEXAGON

exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 1.7596 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 192.7096 mm

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Input, Nozzle Desc: H14"

From: 20

Pressure for Reinforcement Calculations	P	0.3527	MPa
Temperature for Internal Pressure	Temp	150	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	700.00	mm
Design Length of Section	L	2716.6667	mm
Shell Finished (Minimum) Thickness	t	8.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		600.00	mm
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User Entered Minimum Design Metal Temperature		11.40	°C
---	--	-------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		4.0000	in.

Size and Thickness Basis		Minimum	
Nominal Thickness		80	

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	127.3240	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	8.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

This is a Manway, Coupling or Access Opening.

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: H14"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation		3.910	in.
Actual Thickness Used in Calculation		0.295	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.35 \cdot 352 / (138 \cdot 1 - 0.6 \cdot 0.35)$
 $= 0.8998$ mm



Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.35 \cdot 51.2 / (118 \cdot 1 - 0.6 \cdot 0.35)$
 $= 0.1532 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3845 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 204.6407 mm
 Parallel to Vessel Wall, opening length d 102.3204 mm
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 14.9746 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	93.630	207.832	NA
Area in Shell	A1	563.292	252.005	NA
Area in Nozzle Wall	A2	149.468	143.544	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	85.507	85.507	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	798.267	481.056	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= 0.5 (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F (1 - fr1))$ per UG-37(d)
 $= 0.5 (102 \cdot 3.99 \cdot 1 + 2 \cdot 5.99 \cdot 3.99 \cdot 1 (1 - 0.86))$
 $= 207.832 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 102 (1 \cdot 6.5 - 1 \cdot 3.99) - 2 \cdot 5.99$
 $(1 \cdot 6.5 - 1 \cdot 3.99) \cdot (1 - 0.86)$
 $= 252.005 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp) (tn - trn) fr2$
 $= (2 \cdot 15) (5.99 - 0.38) 0.86$
 $= 143.544 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= W_o^2 \cdot fr2 + (W_i - can / 0.707)^2 \cdot fr2$
 $= 10^2 \cdot 0.86 + (0)^2 \cdot 0.86$
 $= 85.507 \text{ mm}^2$



UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 1.8845 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= t_a
 = 1.8845 mm

Available Nozzle Neck Thickness = 7.4898 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 7.49$, $t_r = 0.15$, $c = 1.5 \text{ mm}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.026$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $t_g = 7.49$, $t_r = 0.15$, $c = 1.5 \text{ mm}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.026$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $0.35/5.10 = 0.069$

Weld Size Calculations, Description: H14"

Intermediate Calc. for nozzle/shell Welds $T_{min} = 5.9898 \text{ mm}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.1929 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (208 - 252 + 2 \cdot 5.99 \cdot 0.86 \cdot (1 \cdot 6.5 - 3.99)) \cdot 138)$$

$$= \max(0, -2553.99) \text{ N}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A5 + A4 - (W_i - C_{an} / 707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (144 + 0 + 85.5 - 0 \cdot 0.86) \cdot 138$$

$$= 31605.77 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (144 + 0 + 85.5 + (66.6)) \cdot 138$$

$$= 40793.22 \text{ N}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (144 + 0 + 85.5 + 0 + (66.6)) \cdot 138$$



= 40793.22 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\ &= (3.14/2.0) \cdot 114 \cdot 10 \cdot 0.49 \cdot 118 \\ &= 103800. \text{ N} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.14 \cdot 54.2) \cdot (7.49 - 1.5) \cdot 0.7 \cdot 118 \\ &= 84166. \text{ N} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\ &= (3.14/2.0) \cdot 114 \cdot (8 - 1.5) \cdot 0.74 \cdot 138 \\ &= 119164. \text{ N} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (103800 + 84166) = 187967 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (103800 + 0 + 119164 + 0) = 222965 \text{ N} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (103800 + 119164 + 0) = 222965 \text{ N} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 187966 N	, must exceed W = 0 N	or W1 = 31605 N
Path 2-2 = 222964 N	, must exceed W = 0 N	or W2 = 40793 N
Path 3-3 = 222964 N	, must exceed W = 0 N	or W3 = 40793 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1.66 MPa

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 4.6974 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 140.0214 mm

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Input, Nozzle Desc: H24"

From: 20

Pressure for Reinforcement Calculations	P	0.3500	MPa
Temperature for Internal Pressure	Temp	150	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	700.00	mm
Design Length of Section	L	2716.6667	mm
Shell Finished (Minimum) Thickness	t	8.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		2350.00	mm
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User Entered Minimum Design Metal Temperature		11.40	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		4.0000	in.

Size and Thickness Basis		Minimum	
Nominal Thickness		80	

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	127.3240	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	8.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

This is a Manway, Coupling or Access Opening.

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: H24"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation		3.910	in.
Actual Thickness Used in Calculation		0.295	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.35 \cdot 352 / (138 \cdot 1 - 0.6 \cdot 0.35)$
 $= 0.8929$ mm



Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.35 \cdot 51.2 / (118 \cdot 1 - 0.6 \cdot 0.35)$
 $= 0.1520 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3845 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 204.6407 mm
 Parallel to Vessel Wall, opening length d 102.3204 mm
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 14.9746 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	92.913	207.832	NA
Area in Shell	A1	563.984	252.005	NA
Area in Nozzle Wall	A2	149.498	143.544	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	85.507	85.507	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	798.990	481.056	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= 0.5 (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F (1 - fr1))$ per UG-37(d)
 $= 0.5 (102 \cdot 3.99 \cdot 1 + 2 \cdot 5.99 \cdot 3.99 \cdot 1 (1 - 0.86))$
 $= 207.832 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 102 (1 \cdot 6.5 - 1 \cdot 3.99) - 2 \cdot 5.99$
 $(1 \cdot 6.5 - 1 \cdot 3.99) \cdot (1 - 0.86)$
 $= 252.005 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp) (tn - trn) fr2$
 $= (2 \cdot 15) (5.99 - 0.38) 0.86$
 $= 143.544 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can / 0.707)^2 \cdot fr2$
 $= 10^2 \cdot 0.86 + (0)^2 \cdot 0.86$
 $= 85.507 \text{ mm}^2$



UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 1.8845 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= t_a
 = 1.8845 mm

Available Nozzle Neck Thickness = 7.4898 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 7.49$, $t_r = 0.15$, $c = 1.5 \text{ mm}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.025$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $t_g = 7.49$, $t_r = 0.15$, $c = 1.5 \text{ mm}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.025$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $0.35/5.10 = 0.069$

Weld Size Calculations, Description: H24"

Intermediate Calc. for nozzle/shell Welds $T_{min} = 5.9898 \text{ mm}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.1929 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

= $\max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 = $\max(0, (208 - 252 + 2 \cdot 5.99 \cdot 0.86 \cdot (1 \cdot 6.5 - 3.99)) \cdot 138)$
 = $\max(0, -2553.99) \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

= $(A2+A5+A4-(W_i-Can/.707)^2 \cdot f_{r2}) \cdot S_v$
 = $(144 + 0 + 85.5 - 0 \cdot 0.86) \cdot 138$
 = 31605.77 N

Weld Load [W2]:

= $(A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 = $(144 + 0 + 85.5 + (66.6)) \cdot 138$
 = 40793.22 N

Weld Load [W3]:

= $(A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 = $(144 + 0 + 85.5 + 0 + (66.6)) \cdot 138$



= 40793.22 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\ &= (3.14/2.0) \cdot 114 \cdot 10 \cdot 0.49 \cdot 118 \\ &= 103800. \text{ N} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.14 \cdot 54.2) \cdot (7.49 - 1.5) \cdot 0.7 \cdot 118 \\ &= 84166. \text{ N} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\ &= (3.14/2.0) \cdot 114 \cdot (8 - 1.5) \cdot 0.74 \cdot 138 \\ &= 119164. \text{ N} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (103800 + 84166) = 187967 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (103800 + 0 + 119164 + 0) = 222965 \text{ N} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (103800 + 119164 + 0) = 222965 \text{ N} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 187966 N	, must exceed W = 0 N	or W1 = 31605 N
Path 2-2 = 222964 N	, must exceed W = 0 N	or W2 = 40793 N
Path 3-3 = 222964 N	, must exceed W = 0 N	or W3 = 40793 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1.66 MPa

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 4.6974 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 140.0214 mm

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Input, Nozzle Desc: N42"

From: 20

Pressure for Reinforcement Calculations	P	0.3563	MPa
Temperature for Internal Pressure	Temp	150	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	700.00	mm
Design Length of Section	L	2716.6667	mm
Shell Finished (Minimum) Thickness	t	8.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		200.00	mm
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User Entered Minimum Design Metal Temperature		11.40	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm

Flange Material		SA-105	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	176.6000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	8.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N42"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation		2.000	in.
Actual Thickness Used in Calculation		0.660	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.36 \cdot 352 / (138 \cdot 1 - 0.6 \cdot 0.36)$
 $= 0.9090$ mm



Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= $R(\exp([P/(\text{Sn} \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
= $26.9(\exp([0.36/(138 \cdot 1)] - 1))$
= 0.0695 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.3673 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	107.6000	mm
Parallel to Vessel Wall, opening length	d	53.8000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	16.2500	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N42".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.8673	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 2.4090	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 3.0000	mm
Wall Thickness, shell/head, external pressure	trb2	= 1.7636	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 6.3000	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[6.3, max(3, 3)]
= 3.0000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(1.87, 3)
= 3.0000 mm

Available Nozzle Neck Thickness = 16.7640 mm -> OK

**Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code,
(see 319.4.4 and 302.3.5):**

Sustained	:	10.0,	Allowable	:	138.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	335.0 MPa	Passed
Occasional	:	0.2,	Allowable	:	183.5 MPa	Passed
Shear	:	9.4,	Allowable	:	96.6 MPa	Passed

*The number of cycles on this nozzle was assumed to be 7000 or less for the determination of
the expansion stress allowable.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 8, tr = 0.91, c = 1.5 mm, E* = 1
Thickness Ratio = $tr \cdot E*/(tg - c) = 0.14$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :



Design Pressure/Ambient Rating = 0.36/10.20 = 0.035

Weld Size Calculations, Description: N42"

Intermediate Calc. for nozzle/shell Welds Tmin 6.5000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	4.5500 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 2.5490 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 187.1490 mm

Input Echo, WRC297 Item 1, Description: N42" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	700.000	mm
Shell Thickness	8.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	50.800	mm
Nozzle Thickness	16.7640	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-105	
Nozzle Cold S.I. Allowable	SNmc	138.00 MPa
Nozzle Hot S.I. Allowable	SNmh	138.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	0.36	MPa
Radial Load	P	-1300.00	N
Circumferential Shear	Vc	2000.00	N
Longitudinal Shear	Vl	2000.00	N
Circumferential Moment	Mc	300000.00	N-mm
Longitudinal Moment	Ml	400000.03	N-mm
Torsional Moment	Mt	500000.00	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (5.030) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 = NozzleOD + 2 * 1.65 * $\sqrt{R_{mean}(t - ca)}$
 = 84.328 + 2 * 1.65 * $\sqrt{354.75(8.0 - 1.5)}$
 = 242.792 mm

WRC 297 Curve Access Parameters:



Vessel Mean Diameter (D) = 709.500 mm
 Nozzle Outside Diameter (d) = 84.328 mm
 Vessel Thickness used (T) = 6.500 mm
 Nozzle Thickness used (t) = 15.264 mm
 T / t = 0.426
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.242$$

Nr/P = 0.093
 Mr/P = 0.196
 M0/P = 0.057
 N0/P = 0.118
 MrD/Mc = 0.305
 NrDL/Mc = 0.090
 M0d/Mc = 0.094
 N0DL/Mc = 0.051
 MrD/Ml = 0.168
 NrDL/Ml = 0.086
 M0D/Ml = 0.054
 N0DL/Ml = 0.125

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	10	-10	10	-10
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-14	-14	14	14
Outplane Bending (ML)	-36	36	36	-36
Normal Pressure Stress	19	19	19	19
Outplane Stress Summary	-16	34	83	-9

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	36	-36	36	-36
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-9	-9	9	9
Inplane Bending (ML)	-112	112	112	-112
Inplane Pressure Stress	9	9	9	9
Inplane Stress Summary	-73	79	171	-126

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	2	2	-2	-2
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	6	6	6	6
Shear Stress Summary	9	9	4	4

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	75	81	171	127



Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	10	-10	10	-10
Outplane Membrane (Mc)	-4	-4	4	4
Outplane Bending (Mc)	-47	47	47	-47
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	9	9	9	9
Outplane Stress Summary	-28	45	75	-40

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	36	-36	36	-36
Inplane Membrane (Mc)	-7	-7	7	7
Inplane Bending (Mc)	-153	153	153	-153
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	19	19	19	19
Inplane Stress Summary	-103	132	219	-160

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-2	-2	2	2
Torsional Shear (Mt)	6	6	6	6
Shear Stress Summary	4	4	9	9

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	103	132	220	160

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-14	-14	14	14
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	0	1	0	1
Hoop Stress Summary	-9	-9	18	18

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0



Axial Bending (P)	5	-5	5	-5
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-8	-8	8	8
Axial Bending (ML)	-15	15	15	-15
Axial Pressure Stress	0	0	0	0

Axial Stress Summary	-17	2	29	-11
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	2	2	2	2

Shear Stress Summary	3	3	1	1
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside
Two * Max Shear Stress	19	14	29	30

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-4	-4	4	4
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	0	1	0	1

Hoop Stress Summary	0	0	8	8
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	5	-5	5	-5
Axial Membrane (Mc)	-6	-6	6	6
Axial Bending (Mc)	-23	23	23	-23
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	0	0	0	0

Axial Stress Summary	-24	13	35	-21
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	2	2	2	2

Shear Stress Summary	1	1	3	3
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside
Two * Max Shear Stress	24	13	36	31

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	19	19	19	19	19	19	19	19
Circ. Pl (SUS)	-11	-11	17	17	-5	-5	9	9
Circ. Q (SUS)	-26	26	46	-46	-117	117	189	-189
Long. Pm (SUS)	9	9	9	9	9	9	9	9
Long. Pl (SUS)	-7	-7	11	11	-1	-1	7	7
Long. Q (SUS)	-76	76	148	-148	-37	37	57	-57
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	6	6	6	6	6	6	6	6
Pm (SUS)	19.0	19.0	19.0	19.0	19.0	19.0	19.0	19.0
Pm+Pl (SUS)	8.6	8.6	36.2	36.2	14.6	14.6	28.3	28.3
Pm+Pl+Q (Total)	75.1	79.4	168.2	128.1	103.2	131.2	217.4	161.5

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	19.00	138.00	Passed
Pm+Pl (SUS)	36.25	207.00	Passed
Pm+Pl+Q (TOTAL)	217.44	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	0	1	0	1	0	1	0	1
Circ. Pl (SUS)	-11	-11	17	17	-1	-1	7	7
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	0	0	0	0	0	0	0	0
Long. Pl (SUS)	-8	-8	8	8	-6	-6	6	6
Long. Q (SUS)	-10	10	20	-20	-18	18	28	-28
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0
Pm+Pl (SUS)	11.0	10.0	17.0	18.0	6.0	6.0	7.0	8.0
Pm+Pl+Q (Total)	18.5	12.6	28.4	30.3	24.2	12.6	34.1	30.3

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
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PV Elite 28 Licensee: SL User
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HEXAGON

Pm (SUS)	1.00	138.00	Passed
Pm+Pl (SUS)	18.00	207.00	Passed
Pm+Pl+Q (TOTAL)	34.15	413.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: N24"

From: 30

Pressure for Reinforcement Calculations	P	0.3500	MPa
Temperature for Internal Pressure	Temp	150	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Elliptical Head	D	700.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	8.0000	mm
Head Internal Corrosion Allowance	c	1.5000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	0.0000	mm
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User Entered Minimum Design Metal Temperature		11.40	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		4.0000	in.

Size and Thickness Basis		Minimum	
Nominal Thickness		80	

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	25.6720	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	8.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N24"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation		3.910	in.
Actual Thickness Used in Calculation		0.295	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 0.35 \cdot 703 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 0.35)$
 $= 0.7993$ mm



Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.35 \cdot 51.2 / (118 \cdot 1 - 0.6 \cdot 0.35)$
 $= 0.1520 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.1964 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 204.6407 mm
 Parallel to Vessel Wall, opening length d 102.3204 mm
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 14.9746 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	83.177	96.486	NA
Area in Shell	A1	573.396	467.264	NA
Area in Nozzle Wall	A2	149.498	148.363	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		85.507	85.507	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	808.401	701.134	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= 0.5 (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F (1 - fr1))$ per UG-37(d)
 $= 0.5 (102 \cdot 1.85 \cdot 1 + 2 \cdot 5.99 \cdot 1.85 \cdot 1 (1 - 0.86))$
 $= 96.486 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 102 (1 \cdot 6.5 - 1 \cdot 1.85) - 2 \cdot 5.99$
 $(1 \cdot 6.5 - 1 \cdot 1.85) \cdot (1 - 0.86)$
 $= 467.264 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp) (tn - trn) fr2$
 $= (2 \cdot 15) (5.99 - 0.2) 0.86$
 $= 148.363 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can / 0.707)^2 \cdot fr2$
 $= 10^2 \cdot 0.86 + (0)^2 \cdot 0.86$
 $= 85.507 \text{ mm}^2$



UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 1.6964$ mm
Wall Thickness per UG16(b), $t_{r16b} = 3.0000$ mm
Wall Thickness, shell/head, internal pressure $t_{rb1} = 2.3867$ mm
Wall Thickness $t_{b1} = \max(t_{rb1}, t_{r16b}) = 3.0000$ mm
Wall Thickness, shell/head, external pressure $t_{rb2} = 1.7620$ mm
Wall Thickness $t_{b2} = \max(t_{rb2}, t_{r16b}) = 3.0000$ mm
Wall Thickness per table UG-45 $t_{b3} = 6.7578$ mm

Determine Nozzle Thickness candidate [tb]:

$= \min[t_{b3}, \max(t_{b1}, t_{b2})]$
 $= \min[6.76, \max(3, 3)]$
 $= 3.0000$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(t_a, t_b)$
 $= \max(1.7, 3)$
 $= 3.0000$ mm

Available Nozzle Neck Thickness = 7.4898 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	38.2,	Allowable	:	118.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	256.8 MPa	Passed
Occasional	:	1.4,	Allowable	:	156.9 MPa	Passed
Shear	:	51.5,	Allowable	:	82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govern. thk, $t_g = 7.49$, $t_r = 0.15$, $c = 1.5$ mm, $E^* = 1$
Thickness Ratio = $t_r * E^* / (t_g - c) = 0.025$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govern. thk, $t_g = 7.49$, $t_r = 0.15$, $c = 1.5$ mm, $E^* = 1$
Thickness Ratio = $t_r * E^* / (t_g - c) = 0.025$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $0.35/5.10 = 0.069$

Weld Size Calculations, Description: N24"

Intermediate Calc. for nozzle/shell Welds $t_{min} = 5.9898$ mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.1929 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)



Weld Load [W]:

$$\begin{aligned} &= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv) \\ &= \max(0, (96.5 - 467 + 2 \cdot 5.99 \cdot 0.86 \cdot \\ &\quad (1 \cdot 6.5 - 1.85)) \cdot 138) \\ &= \max(0, -44595.77) \text{ N} \end{aligned}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned} &= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv \\ &= (148 + 0 + 85.5 - 0 \cdot 0.86) \cdot 138 \\ &= 32270.76 \text{ N} \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned} &= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv \\ &= (148 + 0 + 85.5 + (66.6)) \cdot 138 \\ &= 41458.20 \text{ N} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S \\ &= (148 + 0 + 85.5 + 0 + (66.6)) \cdot 138 \\ &= 41458.20 \text{ N} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\ &= (3.14/2.0) \cdot 114 \cdot 10 \cdot 0.49 \cdot 118 \\ &= 103800. \text{ N} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.14 \cdot 54.2) \cdot (7.49 - 1.5) \cdot 0.7 \cdot 118 \\ &= 84166. \text{ N} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\ &= (3.14/2.0) \cdot 114 \cdot (8 - 1.5) \cdot 0.74 \cdot 138 \\ &= 119164. \text{ N} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (103800 + 84166) = 187967 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (103800 + 0 + 119164 + 0) = 222965 \text{ N} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (103800 + 119164 + 0) = 222965 \text{ N} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 187966 N	, must exceed W = 0 N	or W1 = 32270 N
Path 2-2 = 222964 N	, must exceed W = 0 N	or W2 = 41458 N
Path 3-3 = 222964 N	, must exceed W = 0 N	or W3 = 41458 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1.86 MPa

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 2.5713 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 36.2433 mm

Input Echo, WRC107/537 Item 1, Description: N24" :



Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	1.5000	mm
Vessel Diameter	Dv	1260.000	mm
Vessel Thickness	Tv	8.000	mm
Design Temperature	T1	150.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	MPa
Vessel Hot S.I. Allowable	Smh	138.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	1.5000	mm
Nozzle Diameter	Dn	99.320	mm
Nozzle Thickness	Tn	7.490	mm
Nozzle Material		SA-106 B	
Nozzle UNS Number		K03006	
Nozzle Cold S.I. Allowable	SNmc	118.00	MPa
Nozzle Hot S.I. Allowable	SNmh	118.00	MPa
Design Internal Pressure	Dp	0.350	MPa
Include Pressure Thrust		No	

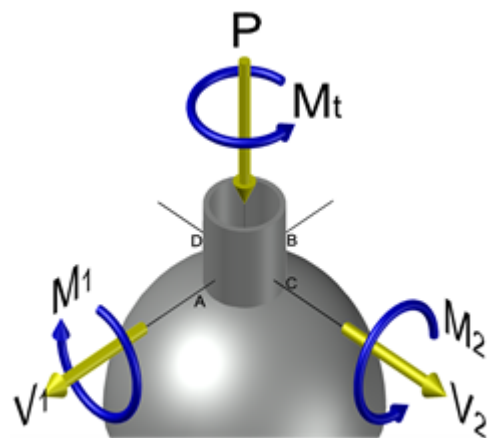
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-2700.0	N
Longitudinal Shear (SUS)	(V1) V1	4000.0	N
Circumferential Shear (SUS)	(Vc) V2	4000.0	N
Circumferential Moment (SUS)	(Mc) M1	1600000.0	N-mm
Longitudinal Moment (SUS)	(Ml) M2	1200000.1	N-mm
Torsional Moment (SUS)	Mt	4500000.5	N-mm

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".





and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 114.3 + 2 \cdot 1.65 \cdot \sqrt{(634.75 (8.0 - 1.5))}$$

$$= 326.269 \text{ mm}$$

WRC 107 Stress Calculation for SUSTained loads:

Radial Load	P	-2700.0	N
Circumferential Shear	(VC) V2	4000.0	N
Longitudinal Shear	(VL) V1	4000.0	N
Circumferential Moment	(MC) M1	1600000.0	N-mm
Longitudinal Moment	(ML) M2	1200000.1	N-mm
Torsional Moment	MT	4500000.5	N-mm

Dimensionless Parameters: U = 0.89 TAU = 9.04 RHO = 1.09

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SP 2	0.03861	(A,B,C,D)
M(x) / P	SP 2	0.04692	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SM 2	0.05816	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SM 2	0.08559	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SM 2	0.05816	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SM 2	0.08559	(A,B,C,D)
N(y) * T / P	SP 2	0.10142	(A,B,C,D)
M(y) / P	SP 2	0.01680	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SM 2	0.10027	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SM 2	0.03734	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SM 2	0.10027	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SM 2	0.03734	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Membr. P	2.5	2.5	2.5	2.5	2.5	2.5	2.5	2.5
Rad.	Bend. P	18.0	-18.0	18.0	-18.0	18.0	-18.0	18.0	-18.0
Rad.	Membr. MC	0.0	0.0	0.0	0.0	-34.3	-34.3	34.3	34.3
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-302.7	302.7	302.7	-302.7
Rad.	Membr. ML	-25.7	-25.7	25.7	25.7	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-227.0	227.0	227.0	-227.0	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-232.2	185.8	273.2	-216.8	-316.5	252.9	357.4	-283.9
Tang.	Membr. P	6.5	6.5	6.5	6.5	6.5	6.5	6.5	6.5
Tang.	Bend. P	6.4	-6.4	6.4	-6.4	6.4	-6.4	6.4	-6.4
Tang.	Membr. MC	0.0	0.0	0.0	0.0	-59.1	-59.1	59.1	59.1
Tang.	Bend. MC	0.0	0.0	0.0	0.0	-132.1	132.1	132.1	-132.1
Tang.	Membr. ML	-44.3	-44.3	44.3	44.3	0.0	0.0	0.0	0.0
Tang.	Bend. ML	-99.0	99.0	99.0	-99.0	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-130.4	54.8	156.3	-54.7	-178.2	73.0	204.1	-72.9
Shear VC		3.4	3.4	-3.4	-3.4	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-3.4	-3.4	3.4	3.4
Shear MT		33.7	33.7	33.7	33.7	33.7	33.7	33.7	33.7
Tot. Shear		37.2	37.2	30.3	30.3	30.3	30.3	37.2	37.2
Str. Int.		244.4	195.6	280.5	222.3	322.8	257.8	365.9	290.3



WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	17.0	17.0	17.0	17.0	17.0	17.0	17.0	17.0
Rad.	Pl (SUS)	-23.2	-23.2	28.2	28.2	-31.8	-31.8	36.7	36.7
Rad.	Q (SUS)	-209.0	209.0	245.0	-245.0	-284.7	284.7	320.6	-320.6
Long.	Pm (SUS)	17.0	17.0	17.0	17.0	17.0	17.0	17.0	17.0
Long.	Pl (SUS)	-37.8	-37.8	50.8	50.8	-52.6	-52.6	65.6	65.6
Long.	Q (SUS)	-92.6	92.6	105.5	-105.5	-125.6	125.6	138.5	-138.5
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	3.4	3.4	-3.4	-3.4	-3.4	-3.4	3.4	3.4
Shear	Q (SUS)	33.7	33.7	33.7	33.7	33.7	33.7	33.7	33.7
Pm (SUS)		17.0	17.0	17.0	17.0	17.0	17.0	17.0	17.0
Pm+Pl (SUS)		21.6	21.6	68.3	68.3	36.2	36.2	83.0	83.0
Pm+Pl+Q (Total)		227.4	212.6	297.5	205.3	305.8	274.8	382.9	273.2

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	17.00	138.00	Passed
Pm+Pl (SUS)	82.98	207.00	Passed
Pm+Pl+Q (TOTAL)	382.92	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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